Key words And Terms

white light	白光
monochromatic	单色光
coherent light	相干光
coherent source	相干光源
interference	干涉
optical path (length)	光程
optical path (length) difference	光程差
bright (dark) fringe	明（暗）条纹
Young’s Double-slit experiment	杨氏双缝实验

film interference	**薄膜干涉**
incident (reflected) light	**入(反)射光**
half-wavelength loss	**半波损失**
parallel plane thin film	**平行平面薄膜**
wedge-shaped thin film	**劈尖**
Newton's Rings	**牛顿环**
transmission enhanced film	**增透膜**
refraction law	**折射定律**
index of refraction	**折射率**
Michelson interferometer	**迈克耳孙干涉仪**


100

Optics

Geometric optics :reflection , refraction

Wave optics :interference, diffraction, polarization

Quantum optics: interaction between light and matter

主流线索

光的微粒说 **I Newton** *Opticks* 1704

光的波动说 **C Huygens** *Traite de Lumiere* 1690

干涉现象 **T Young** 1801

偏振现象 **E L Malus** 1808

衍射现象 **A J Fresnel** 1816

电磁理论 **J C Maxwell** 1865

光量子、光子理论 **M Planck** 1900，**A Einstein** 1905

红宝石激光 **T H Maiman 1960**

Chapter 18 Interference

18.1 Source of light, Interference of light

18.2 Optical path

18.3 Young's Double-slit experiment

18.4 Interference in thin film

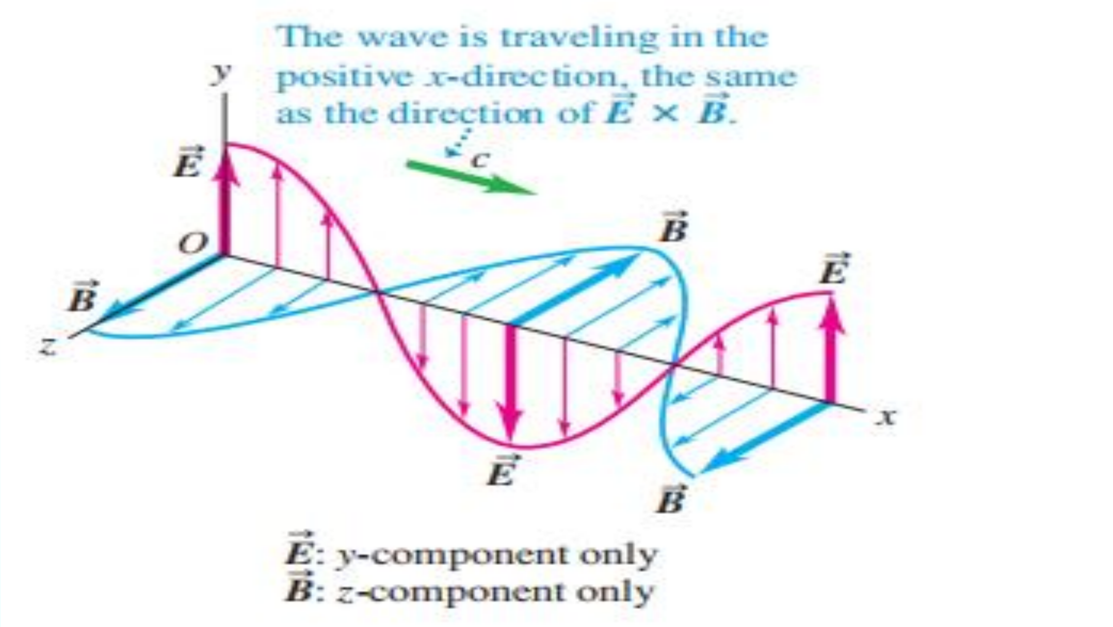
18.5 Michelson interferometer

Chapter 18 *Interference*

18.1 Source of light, Interference of light

Speed in free space

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}$$



Wave equation

$$\begin{cases} E = E_0 \cos \omega (t - \dfrac{r}{v}) \\ B = B_0 \cos \omega (t - \dfrac{r}{v}) \end{cases}$$

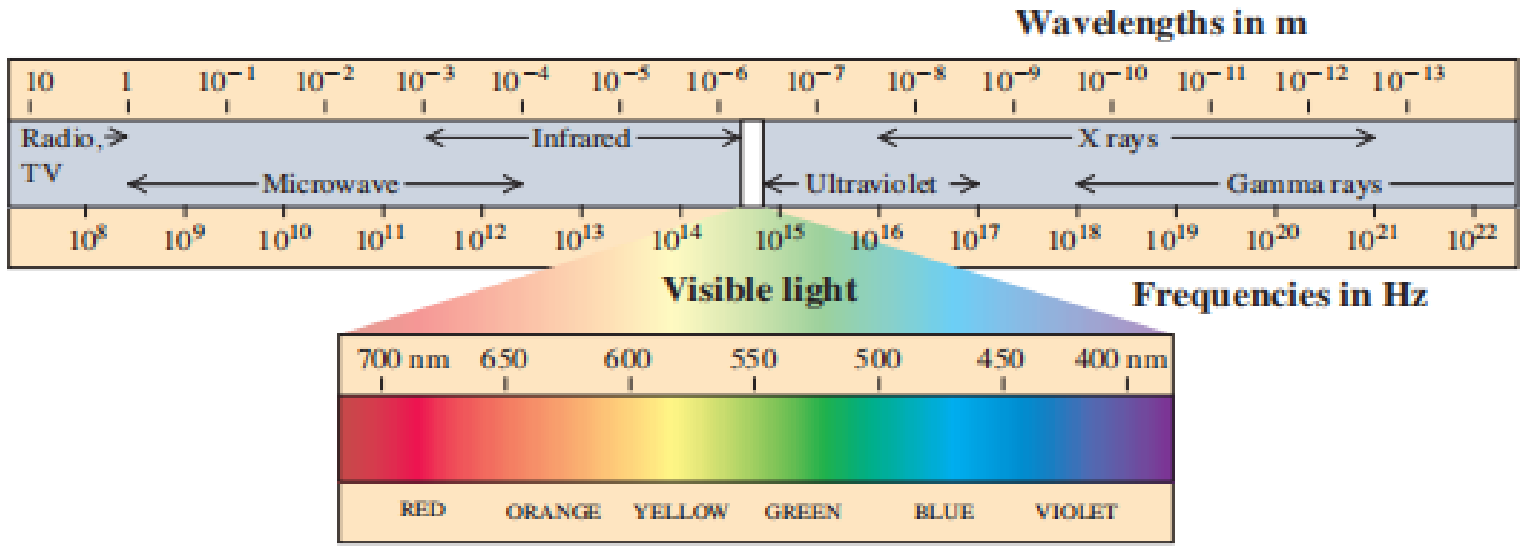
***E*— optical field; The effects of *B* on the human eyes and on various light detectors are exceedingly small.**

机械波　　光波　　概率波

无线电波　　电子双缝干涉

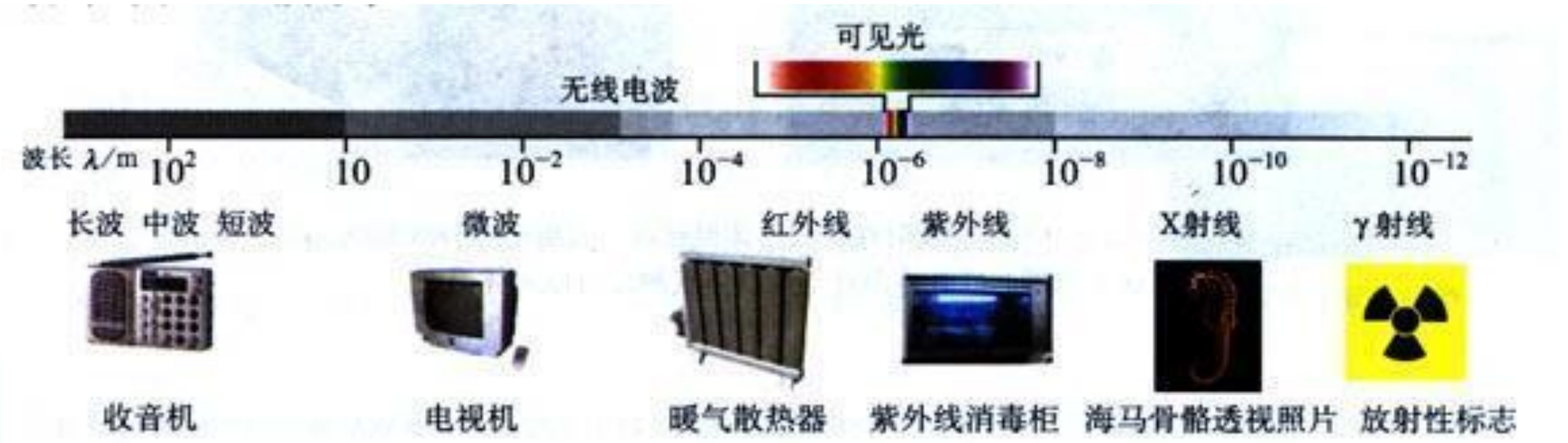
X射线　　量子干涉

Electromagnetic spectrum



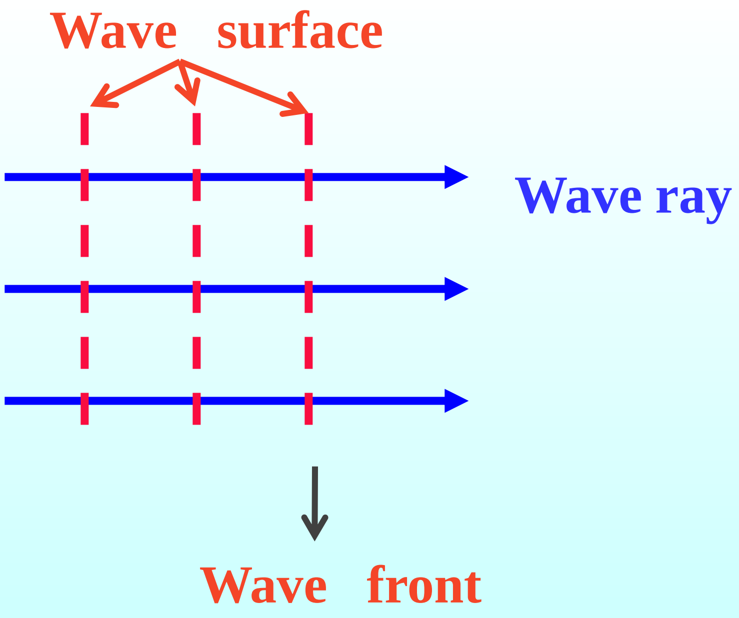
Electromagnetic spectrum

Visible light （400-700nm)

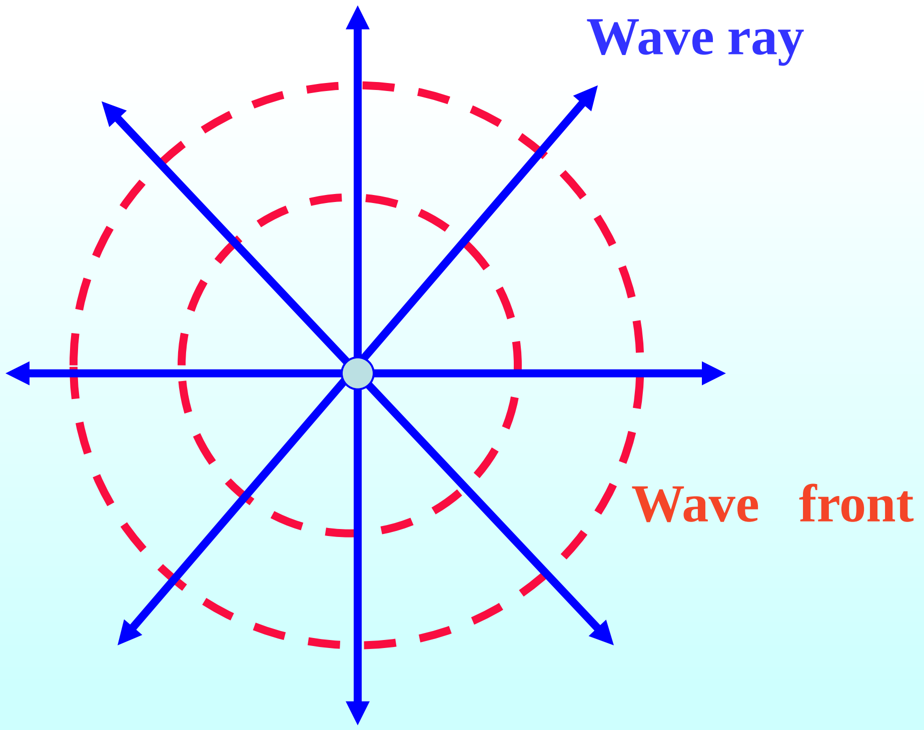


Radio waves **Micro waves** **Infrared** **ultraviolet** **X-rays** **Gamma rays**

plane wave



spherical wave

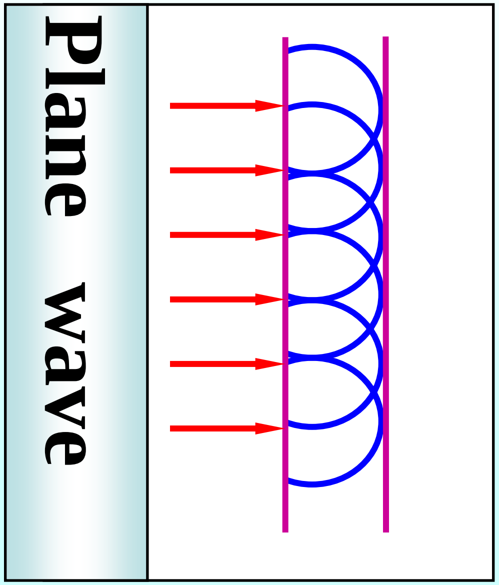


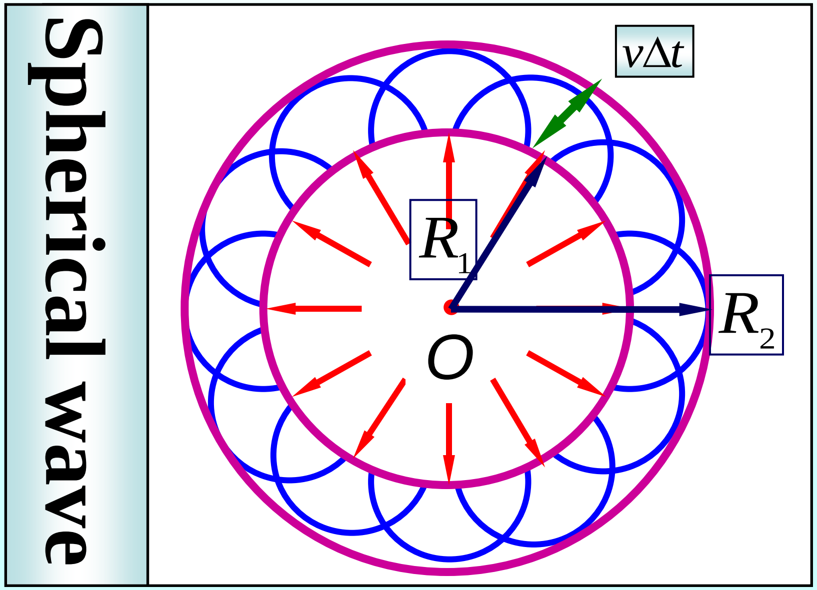
1. Huygens' principle

All points on a wavefront that will severs as point sources of spherical secondary wavelets. After a time *t*, the new position of the wavefront will be that of a surface tangent to these secondary wavelets.

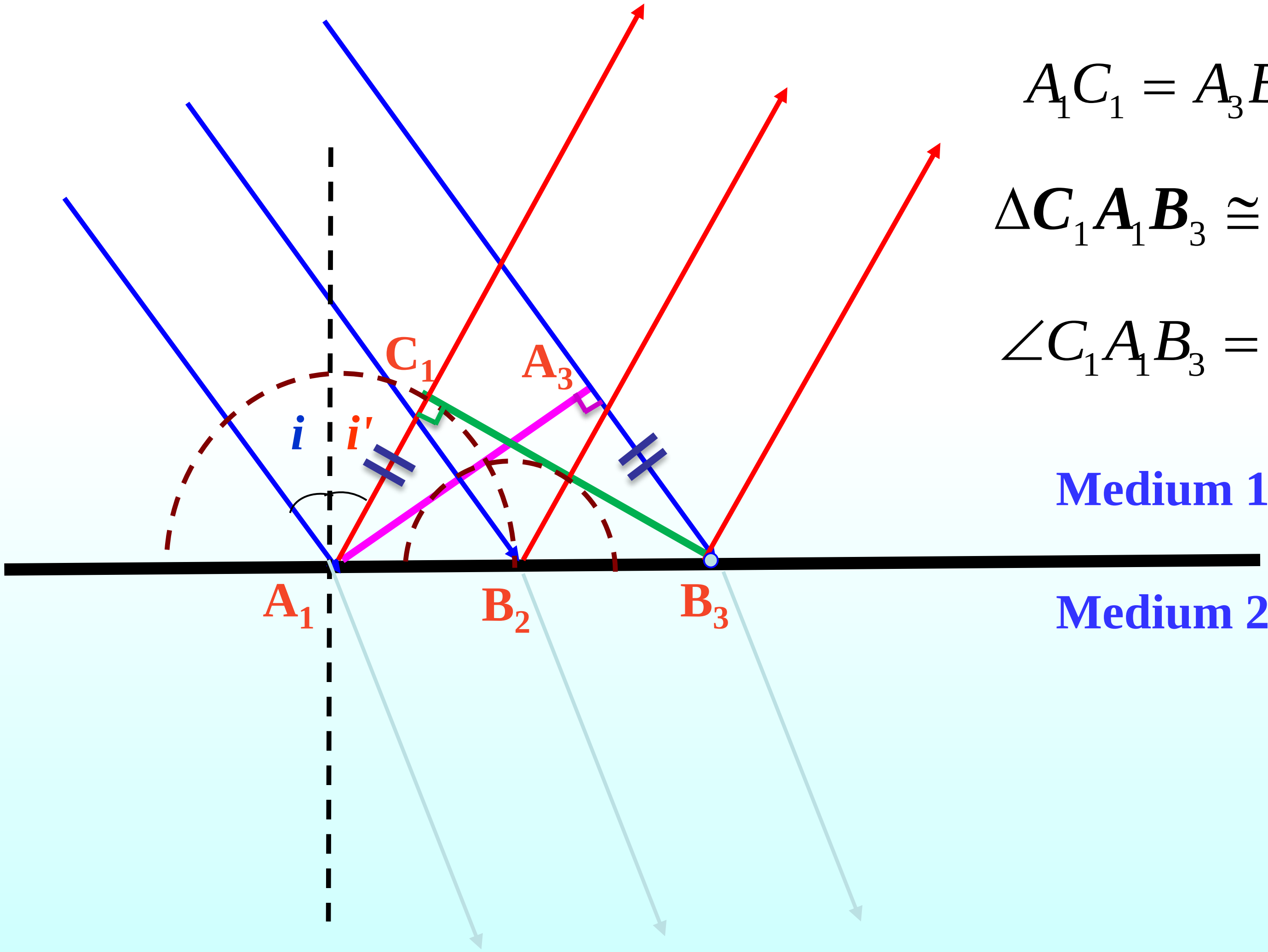
介质中波动传播到的各点都可以看作是发射子波的波源，而在其后的任意时刻，这些子波的包络就是新的波前.

Huygens' wave theory is based on a geometrical construction that allows us to tell where a given wavefront will be if we knows its present position.





Reflection explained, using Huygens' principle



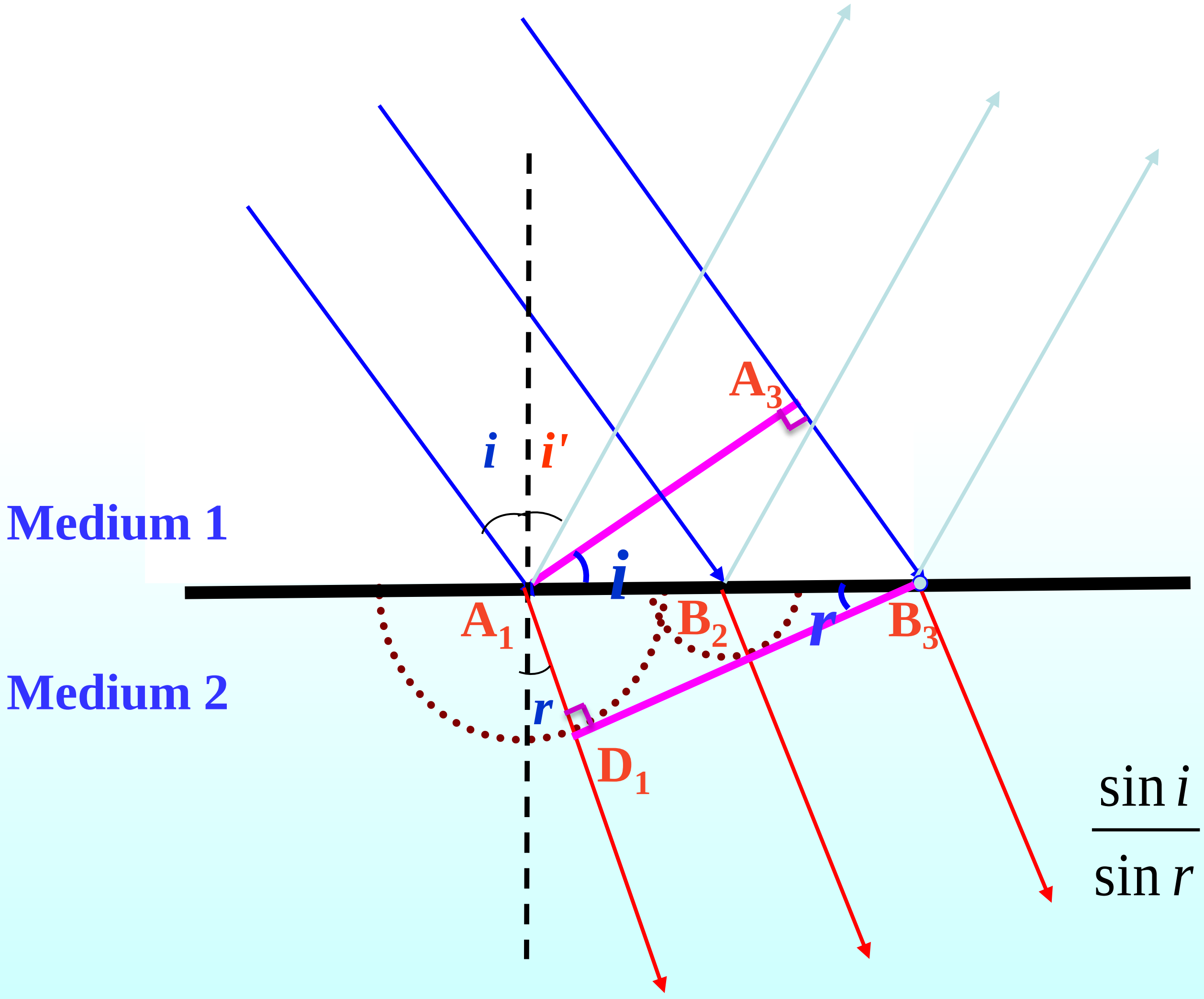
$$A_1C_1 = A_3B_3 = v\Delta t$$

$$\Delta \boldsymbol{C_1A_1B_3} \cong \Delta \boldsymbol{A_3B_3A_1}$$

$$\angle C_1A_1B_3 = \angle A_3B_3A_1$$

$$\boxed{i = i'}$$

Refraction explained, using Huygens' principle



$$A_3B_3 = v_1 \Delta t$$

$$A_1D_1 = v_2 \Delta t$$

$$\sin i = \frac{A_3B_3}{A_1B_3}$$

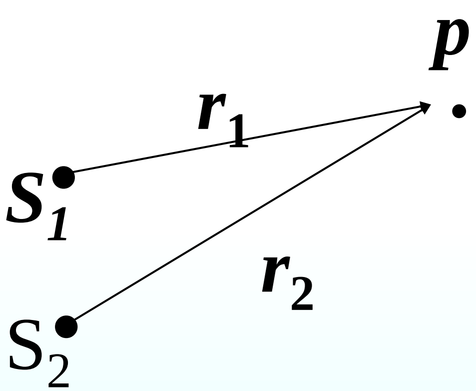
$$\sin r = \frac{A_1D_1}{A_1B_3}$$

$$\frac{\sin i}{\sin r} = \frac{A_3B_3}{A_1D_1} = \frac{v_1 \Delta t}{v_2 \Delta t} = \frac{v_1}{v_2} = \frac{n_2}{n_1}$$

$$n_1 \sin i = n_2 \sin r$$

2. Interference of light

$$y = A\cos(\omega t - kx + \varphi)$$



$$y_1 = A\cos(\omega t - kr_1 + \varphi_1)$$

$$y_2 = A\cos(\omega t - kr_2 + \varphi_2)$$

$$\Delta\varphi = (\varphi_2 - \varphi_1) - 2\pi\frac{r_2 - r_1}{\lambda} = \pm 2k\pi$$

in phase

$$A = A_1 + A_2$$ **Constructive**

$$I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2} \qquad I \propto A^2$$

$$\Delta\varphi = (\varphi_2 - \varphi_1) - 2\pi \frac{r_2 - r_1}{\lambda} = \pm(2k+1)\pi \qquad (k = 0,1,2,\cdots)$$

out of phase

$$A = \left| A_1 - A_2 \right|$$

Destructive

$$I_{\max} = I_1 + I_2 - 2\sqrt{I_1 I_2}$$

$$if \quad \varphi_2 = \varphi_1$$

$$\delta = r_2 - r_1 (\text{path difference})$$

$$\Delta\varphi = \frac{\delta}{\lambda} 2\pi$$

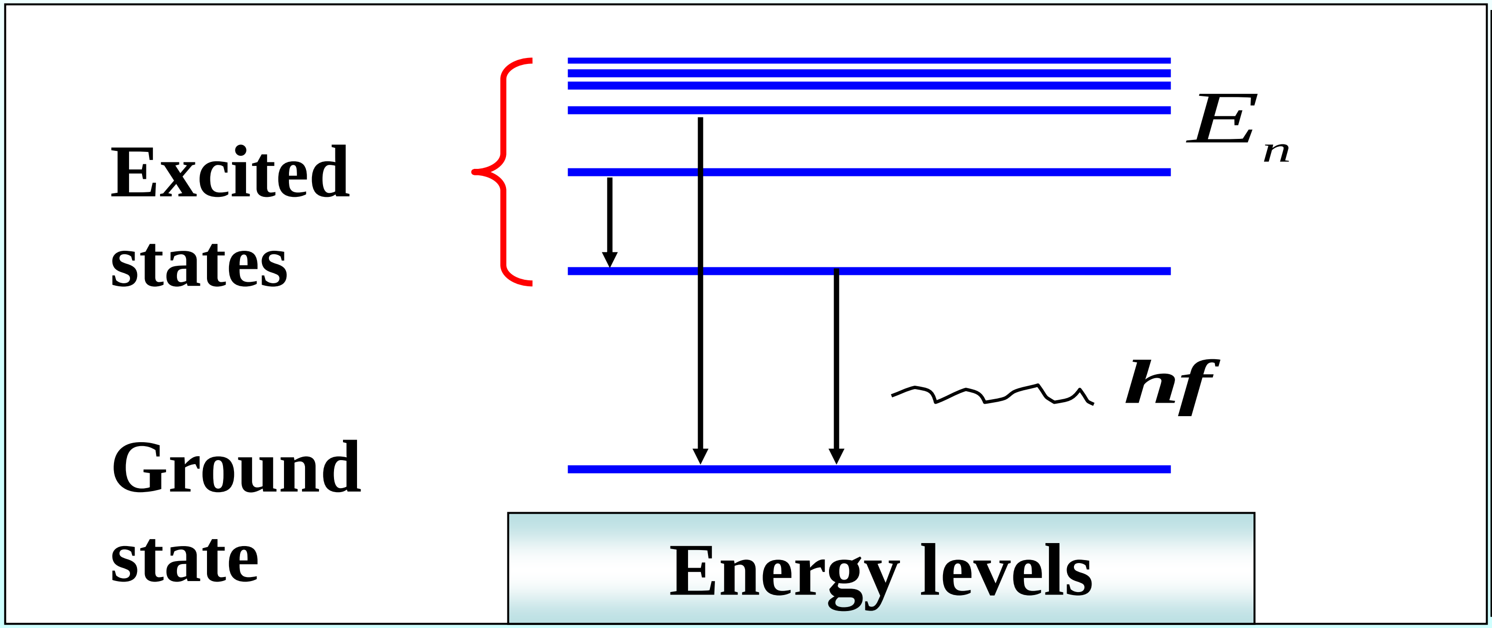
3. Source and coherence

Ordinary source: incoherent sources (phase of waves have no fixed relationship to each other).

Laser source: coherent sources (maintain a constant with respect to each other over time)

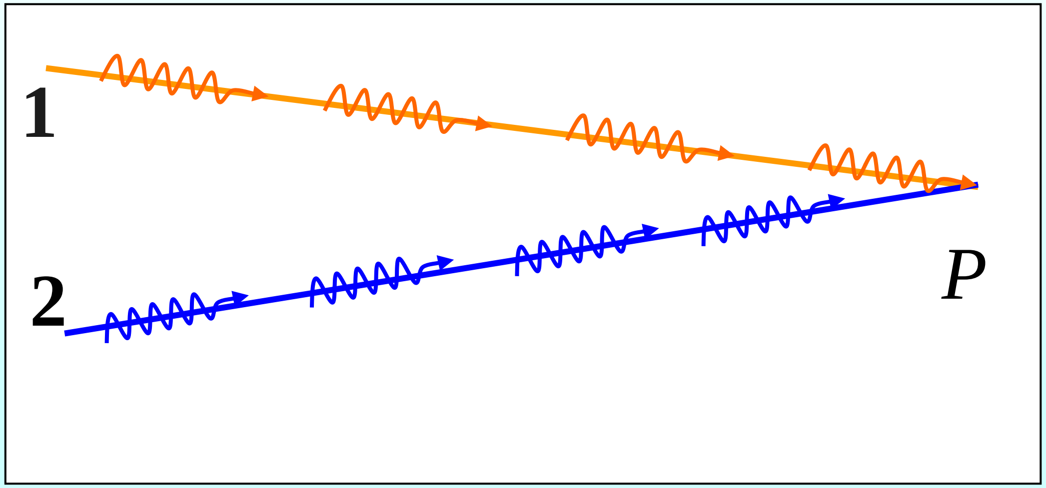
Monochromatic light : wavelength has a certain value.

white light : wavelengths are in a certain range



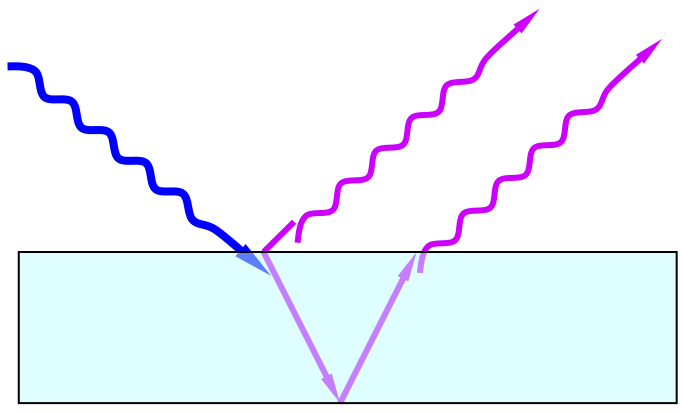
Coherence source

1. The same vibrating direction
2. Same frequency
3. Stable phase difference

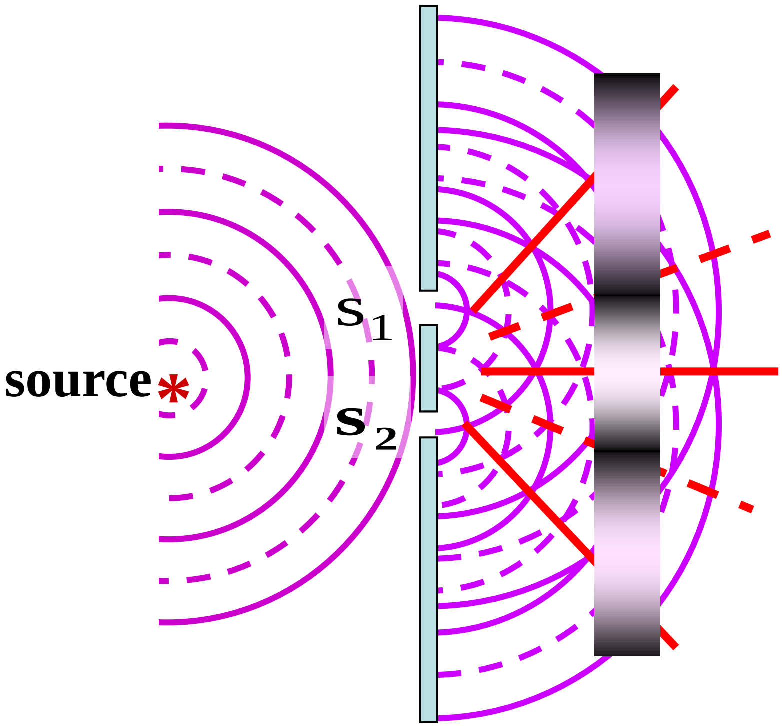


Two ways of getting coherent light

Amplitude-splitting分振幅



Wave front-splitting分波前

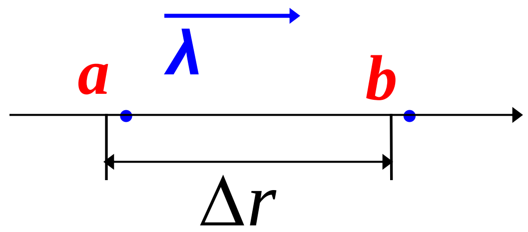


film interference
Michelson interferometer

Young's experiment,
Lloyd mirror

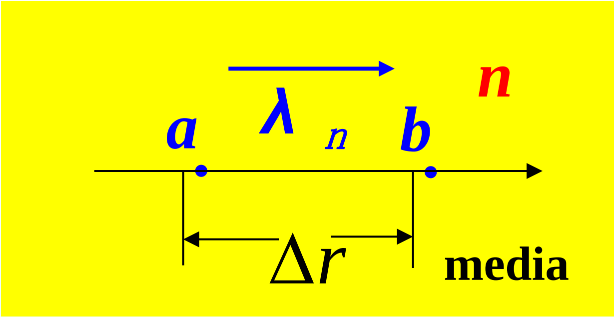
18.2 Optical path (length)

Speed in free space $c = 1\big/\sqrt{\varepsilon_0 \mu_0}$



Speed in the medium $v = 1\big/\sqrt{\varepsilon \mu}$

$$\boxed{v = \frac{c}{n},\ f_n = f}$$



The frequency of the light in the medium is the same as it is in vacuum.

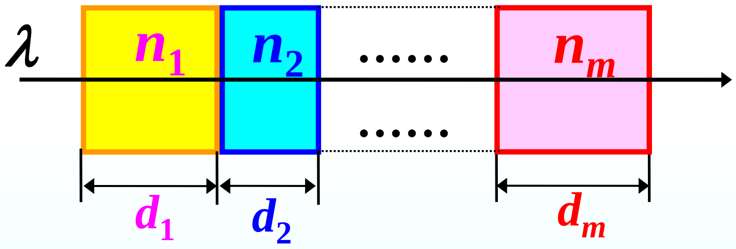
λ_n **: Wavelength in media**

$$\lambda_n = \frac{v}{f} = \frac{c/n}{f} = \frac{c/f}{n} = \frac{\lambda}{n}$$

In free space, the phase difference $\Delta\varphi$ due to path Δr is

$$\Delta\varphi = k\Delta r = \frac{2\pi}{\lambda}\Delta r$$

In a media we may write



$$\Delta\varphi' = \frac{2\pi}{\lambda_n}\Delta r = n\frac{2\pi}{\lambda}\Delta r$$

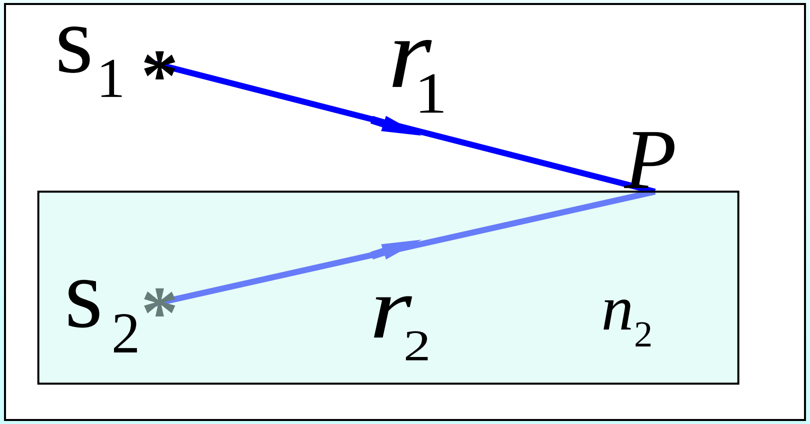
That means the phase difference $\Delta\varphi$ due to path Δr in a media is as same as in free space caused by the path $n\Delta r$. We call $n\Delta r$ the optical path.

光程：折射率*n*和几何路程 $L(L=\Delta r)$ 的乘积。

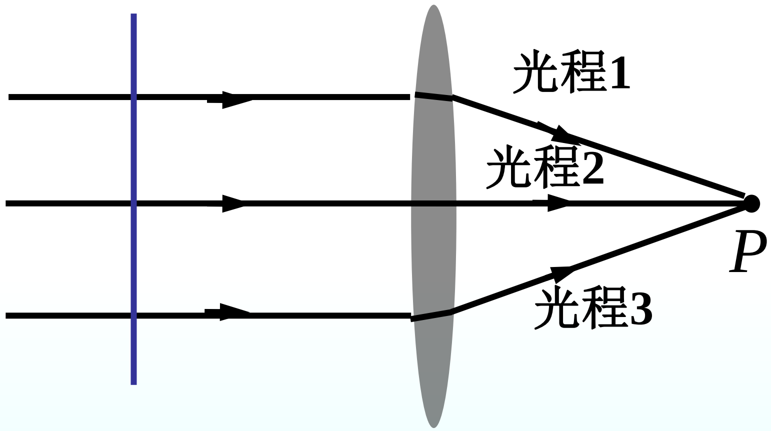
optical path difference

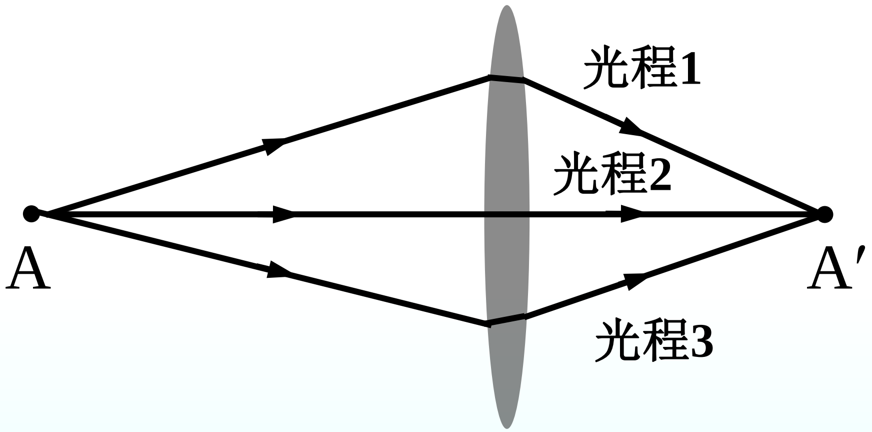
optical path difference $\Delta = nr_2 - r_1$

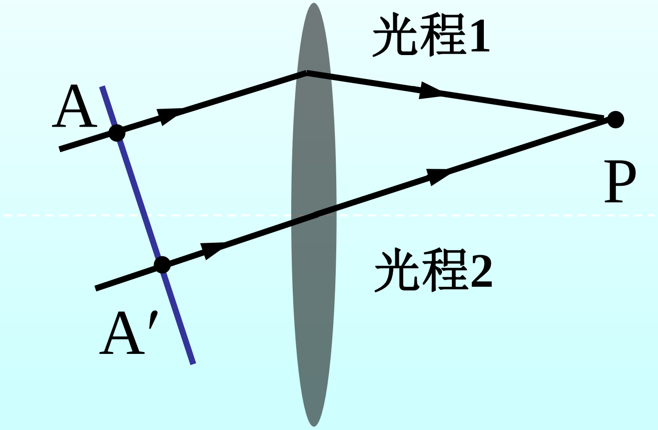
phase difference $\Delta\varphi = 2\pi \dfrac{\Delta}{\lambda}$



It is proved that for near axial rays a thin lens will not cause additional optical path difference.

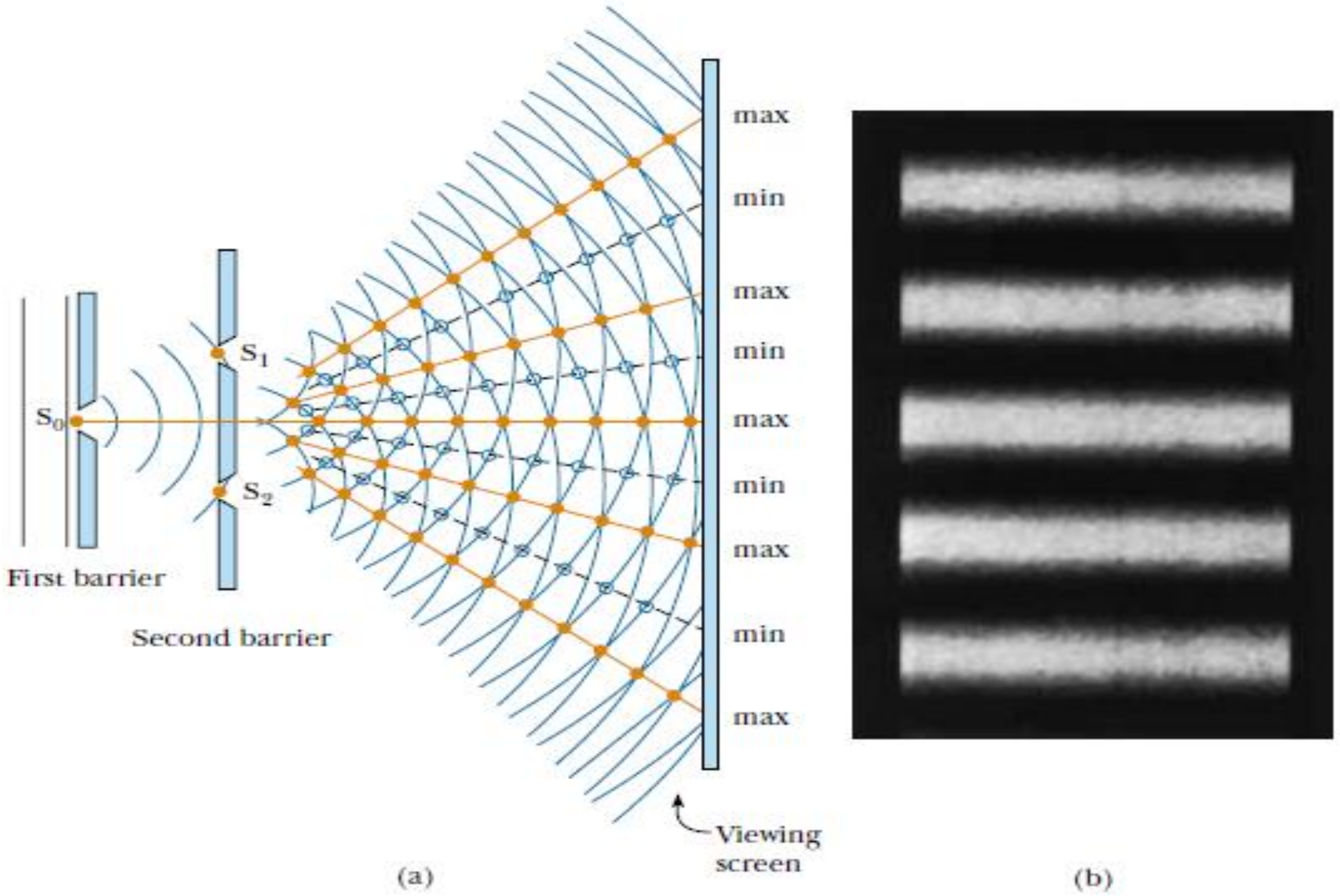




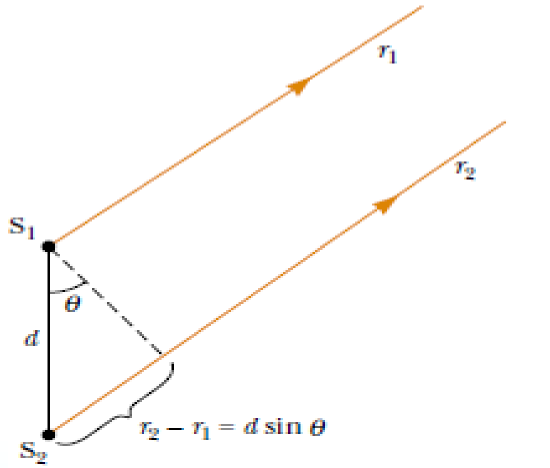


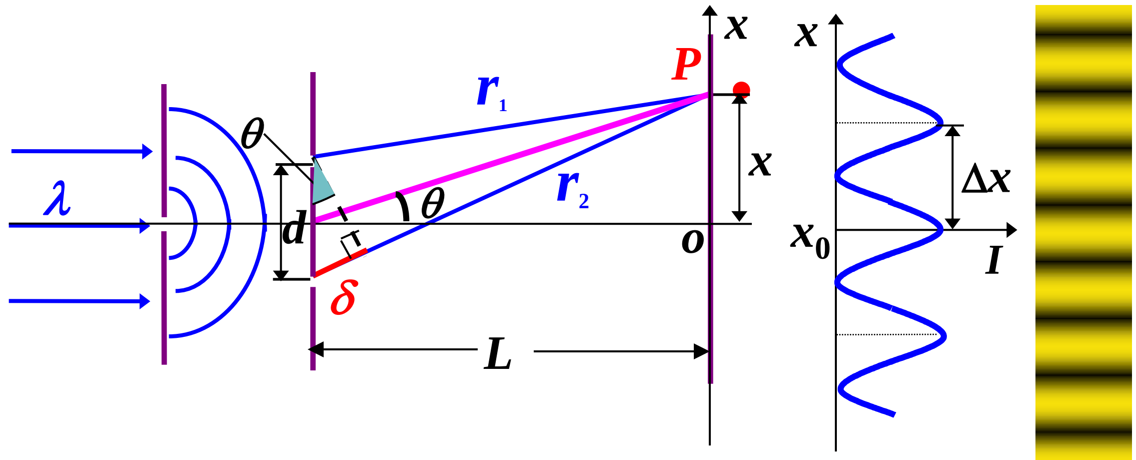
path1=path2=path3

18.3 Young’s Double-slit experiment



$d >> \lambda$, $L >> d$ parallel lights





Wave path difference $\delta = r_2 - r_1 \approx d\sin\theta$

L >> x*：*sinθ ≈ tanθ

$$\boxed{\delta = d\sin\theta \approx d\,\mathrm{tg}\,\theta = d\cdot\frac{x}{L}}$$

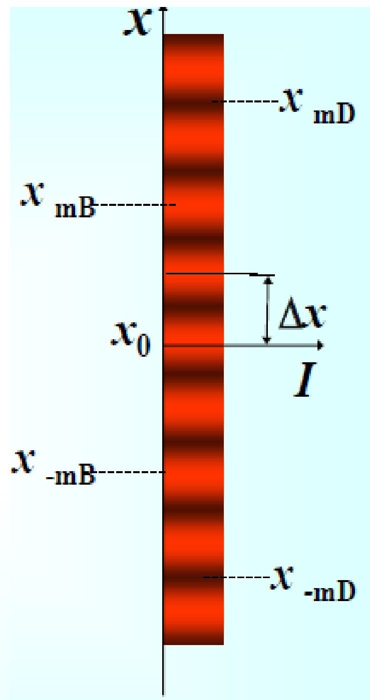
$$\frac{d}{L}x = \pm m\lambda, \quad m = 0, 1, 2, \cdots$$ **Constructive interference**

$$\frac{d}{L}x = \pm(2m+1)\frac{\lambda}{2}, \quad m = 0,1, 2, \cdots$$ **Destructive interference**

***m*: order number**

The central position of the *m*th bright fringes is

$$x_{mB} = \pm m\frac{L}{d}\lambda, \quad m = 0, 1, 2, \cdots$$

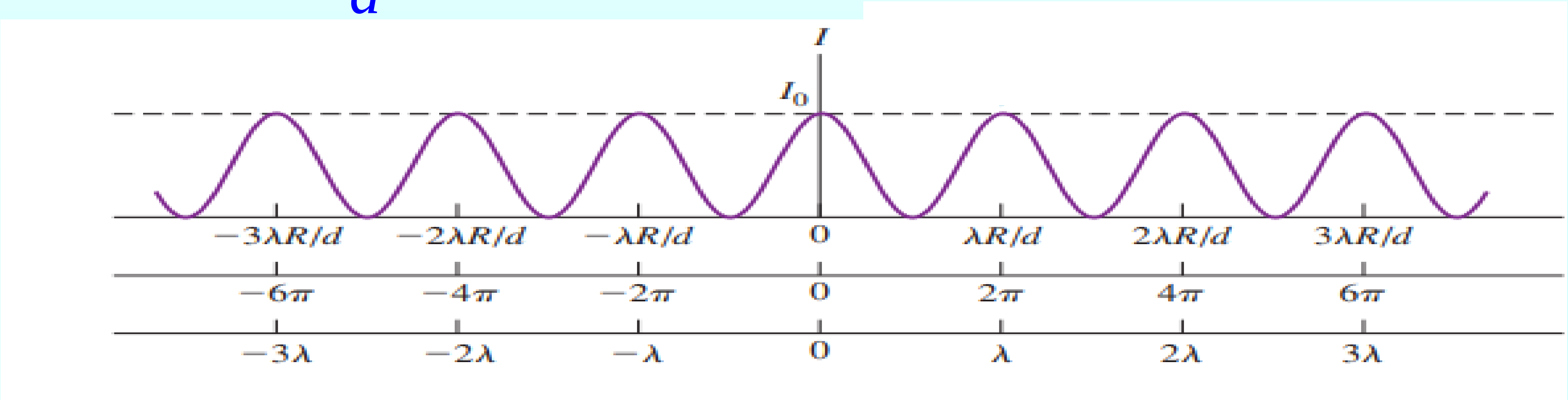


The central positions of the *m*th dark fringes is

$$x_{mD} = \pm\left(m + \frac{1}{2}\right)\frac{L}{d}\lambda, \quad m = 0, 1, 2, \cdots$$

The bright (dark) fringes are equally spaced

$$\Delta x = \frac{L}{d}\lambda$$

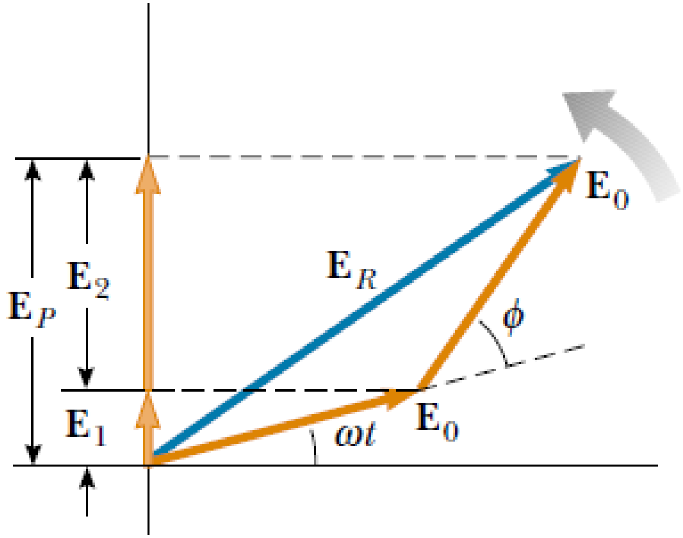


* *Intensity in the double-slit diffraction pattern*

$$E_1 = E_0 \sin \omega t$$

$$E_2 = E_0 \sin(\omega t + \phi)$$

$$\Delta\phi = \frac{2\pi}{\lambda} d \sin\theta$$



At the center $\theta = 0 \;,\quad \Delta\phi = 0$ $\qquad E_R(\theta = 0) = 2E_0$

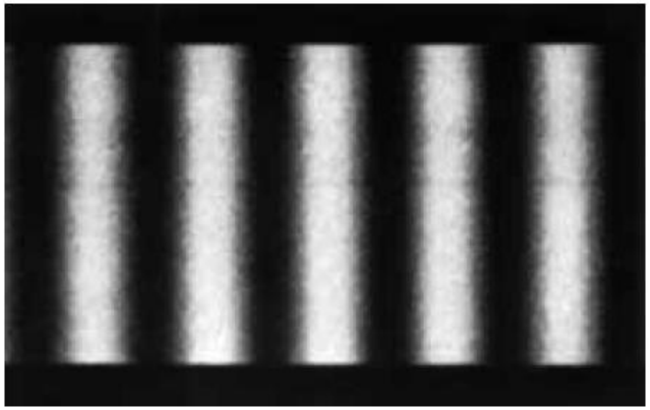
At the point *P* $E_R(\theta) = \sqrt{E_0^2 + E_0^2 + 2E_0E_0 \cos\Delta\varphi} = 2E_0 \cos\frac{\Delta\varphi}{2}$

$$\frac{I_\theta}{I_0} = \frac{(2E_0 \cos\frac{\Delta\varphi}{2})^2}{(2E_0)^2} = \cos^2\frac{\Delta\varphi}{2}$$

$$I_\theta = I_0 \cos^2\frac{\Delta\varphi}{2} = I_0 \cos^2(\frac{\pi d}{\lambda}\sin\theta)$$

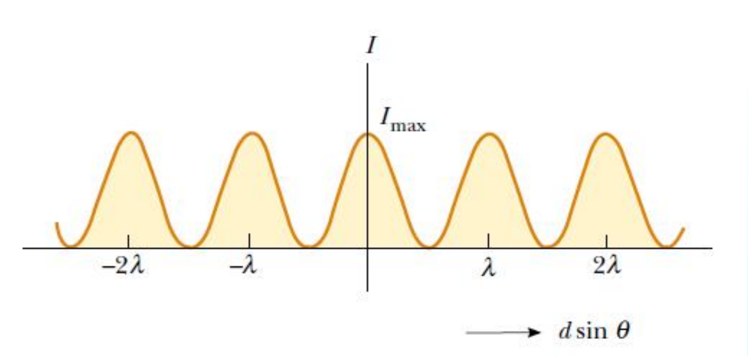
Intensity: uniform $x \ll L,\ \ d \ll L$

$$I_\theta = I_0 \cos^2\left(\frac{\pi d}{\lambda}\sin\theta\right) = I_0 \cos^2\left(\frac{\pi d}{\lambda L}x\right)$$



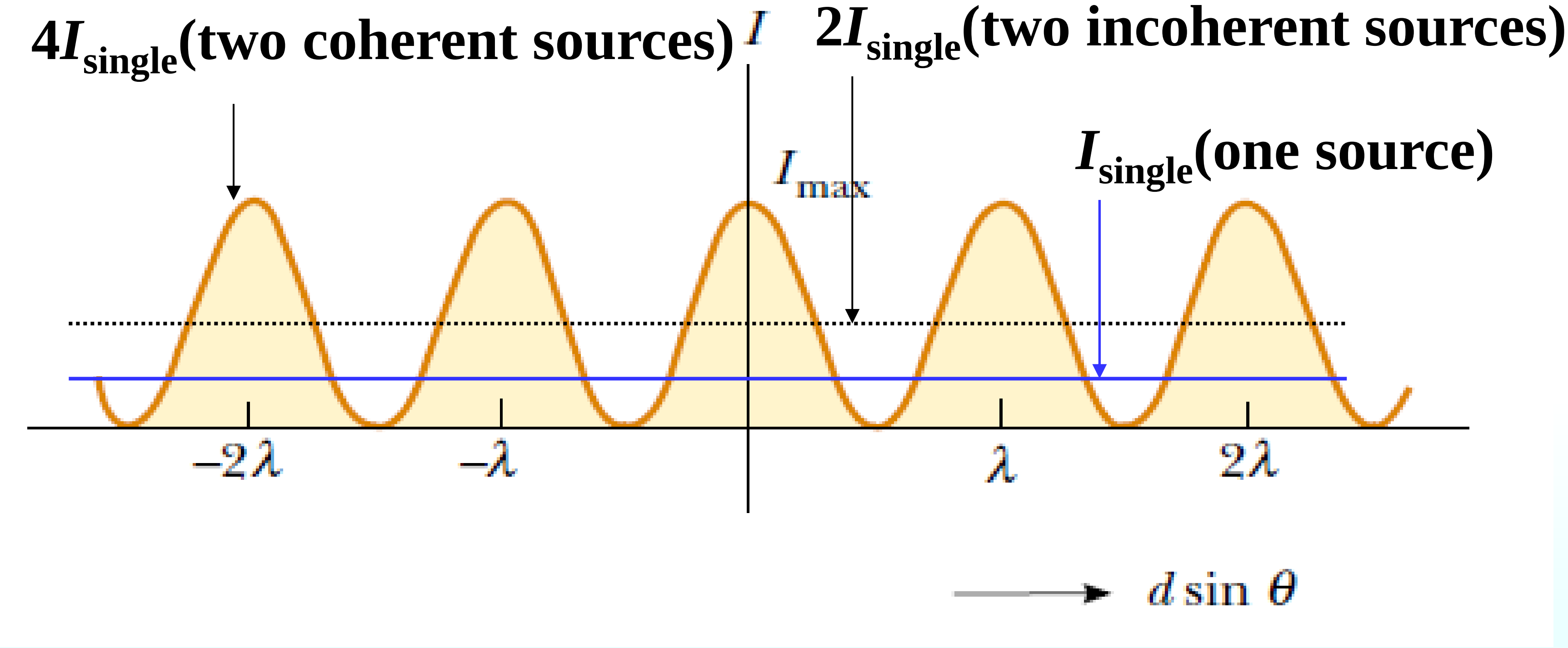
$$\frac{\pi d}{\lambda L}x = m\pi \qquad \text{maxima}$$

$$x_{mB} = \pm m\frac{L}{d}\lambda, \quad m = 0, 1, 2, \cdots$$



$$\frac{\pi d}{\lambda L}x = (m+\frac{1}{2})\pi \qquad \text{mixima}$$

$$x_{mB} = \pm m\frac{L}{d}\lambda, \quad m = 0, 1, 2, \cdots$$

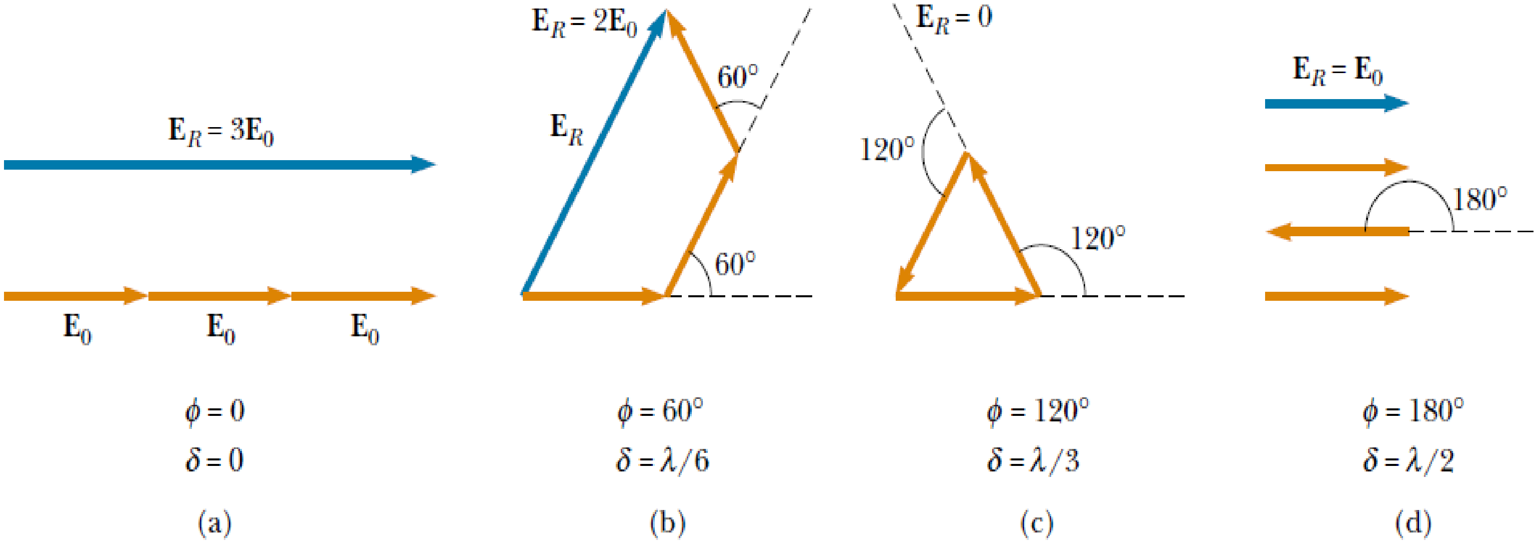


Coherent $I_{\max} = 4I_{\sin gle}$ $I_{\min} = 0$

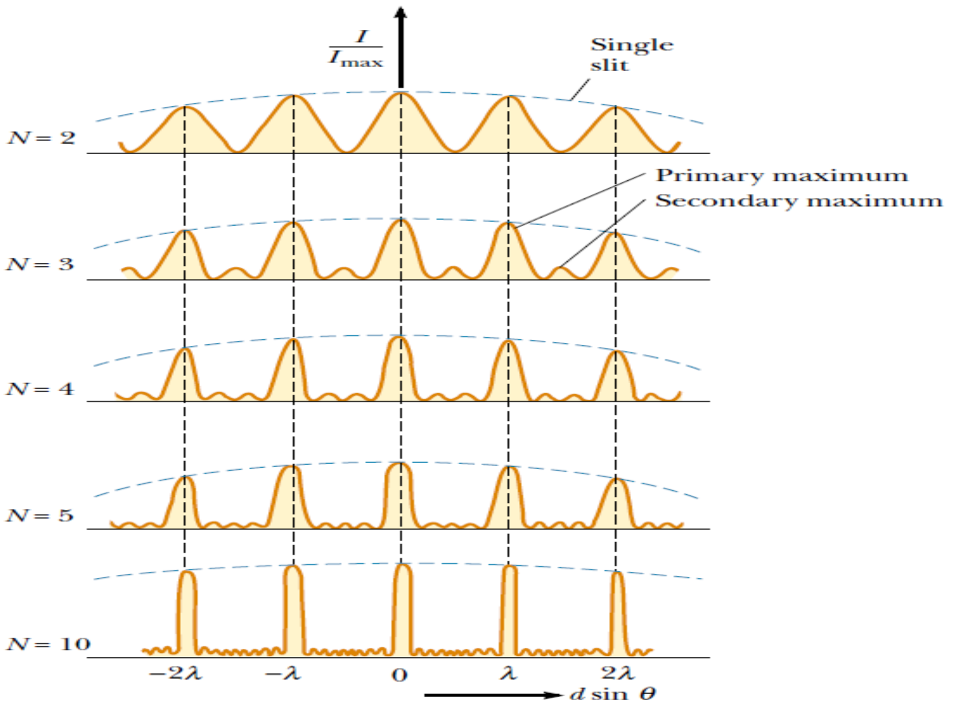
Incoherent $I = 2I_{\sin gle}$

Interference cannot create or destroy energy but merely redistributes it over the screen. The average intensity must be the same regardless of whether the sources are coherent.

Three-slit diffraction pattern



Multiple-slit diffraction pattern



Discussion:

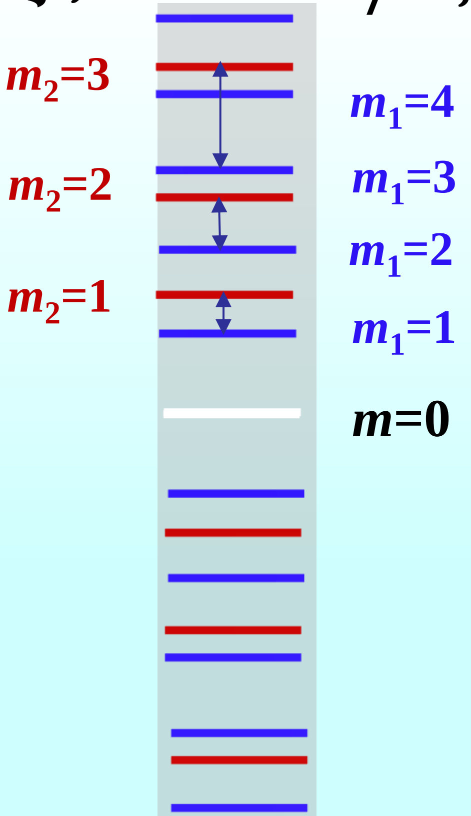
$$\boxed{\Delta x = \frac{L}{d}\lambda}$$

(1) 一系列平行的明暗相间的等间距条纹；

(2) 中央条纹是明纹；

(3) 当L和d不变时 $\Delta x \propto \lambda$，当L和λ不变时，$\Delta x \propto 1/d$；

(4) 白光入射，中央仍是明纹，其余各级次按波长不同排成光谱.



***Question*：** **One of the two slits in a Young's experiment is painted over so that it transmits only one-half the intensity of the other slit. As a result:**

A. the fringe system disappears

B. the bright fringes get brighter and the dark ones get darker

C. the fringes just get dimmer

D. the dark fringes just get brighter

E. the dark fringes get brighter and the bright ones get darker

***Question*：** **light wave with an electric field amplitude of $2E_0$ and a phase constant of zero is to be combined with one of the following waves:**

(1) wave A has an amplitude of E_0 and a phase constant of zero
(2) wave B has an amplitude of E_0 and a phase constant of π
(3)wave C has an amplitude of $2E_0$ and a phase constant of zero
(4)wave D has an amplitude of $2E_0$ and a phase constant of π
(5) wave E has an amplitude of $3E_0$ and a phase constant of π

Which of these combinations produces the least intensity?

1. **Known** $\lambda = 589.3$ nm, L=800 mm

Find Δx **for** (1)d = 1.0 mm; (2)d = 10 mm

Solution:

(1)

$$\Delta x = \frac{L}{d}\lambda$$

$$= \frac{800\,\text{mm}}{1.0\,\text{mm}} \times 589.3 \times 10^{-6}\,\text{mm}$$

$$= 0.47\,\text{mm}$$

(2)

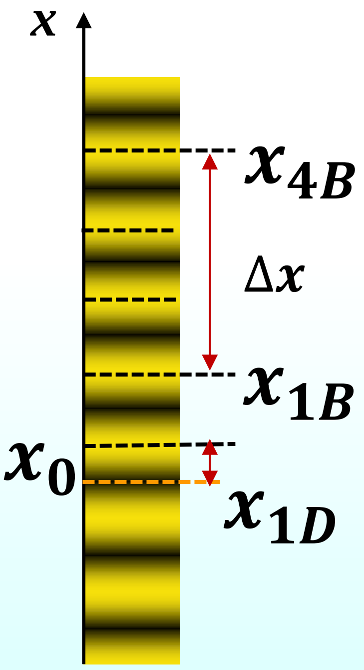
$$\Delta x = 0.047\,\text{mm}$$

2. **Known** d = 0.20 nm, L = 1.0 m

(1) **Find** λ **for bright fringe** $x_4 - x_1 = 7.5\text{mm}$

(2) Δx, **known** λ = 600 nm

Solution:



(1) $$\Delta x_{1\to4} = x_4 - x_1 = \frac{L}{d}(m_4 - m_1)\lambda$$

$$\lambda = \frac{d}{L}\frac{\Delta x_{1\to4}}{(k_4 - k_1)} = \frac{0.20\,\text{mm}}{1.0\times10^3\text{mm}}\frac{7.5\,\text{mm}}{4-1} = 500\ \ \text{nm}$$

(2) $$\Delta x = \frac{L}{d}\lambda = \frac{1.0\times10^3\text{mm}}{0.20\ \ \text{mm}}\times600\times10^{-6}\text{mm}$$

$$=3.0\text{mm}$$

3. **White light passes through two slits 0.50mm apart and an interference pattern is observed on a screen 2.5m away. The violet light falls about 2.0mm and the red 3.5mm from the center of the central white fringe. Estimate the wavelengths of the violet light and the red light.**

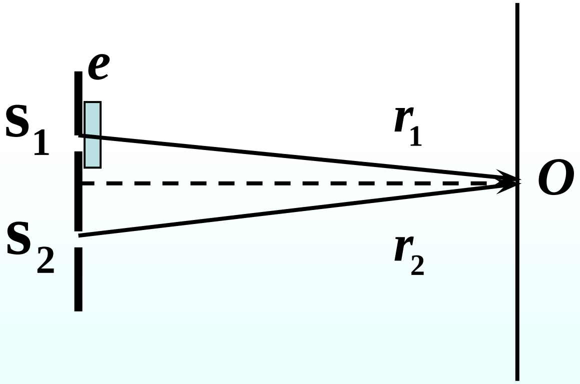
Solution: For violet light

$$\lambda = \frac{d}{L}\frac{x}{m} = \left(\frac{5.0\times10^{-4}}{1}\right)\left(\frac{2.0\times10^{-3}}{2.5}\right) = 400\ \text{nm}$$

For red light

$$\lambda = \frac{d}{L}\frac{x}{m} = \left(\frac{5.0\times10^{-4}}{1}\right)\left(\frac{3.5\times10^{-3}}{2.5}\right) = 700\ \text{nm}$$

4. A very thin sheet of plastic (n=1.60) covers one slit of a double-slit apparatus illuminated by 640-nm light. The center point on the screen, instead of being a maximum, is dark. What is the (minimum) thickness of the plastic?

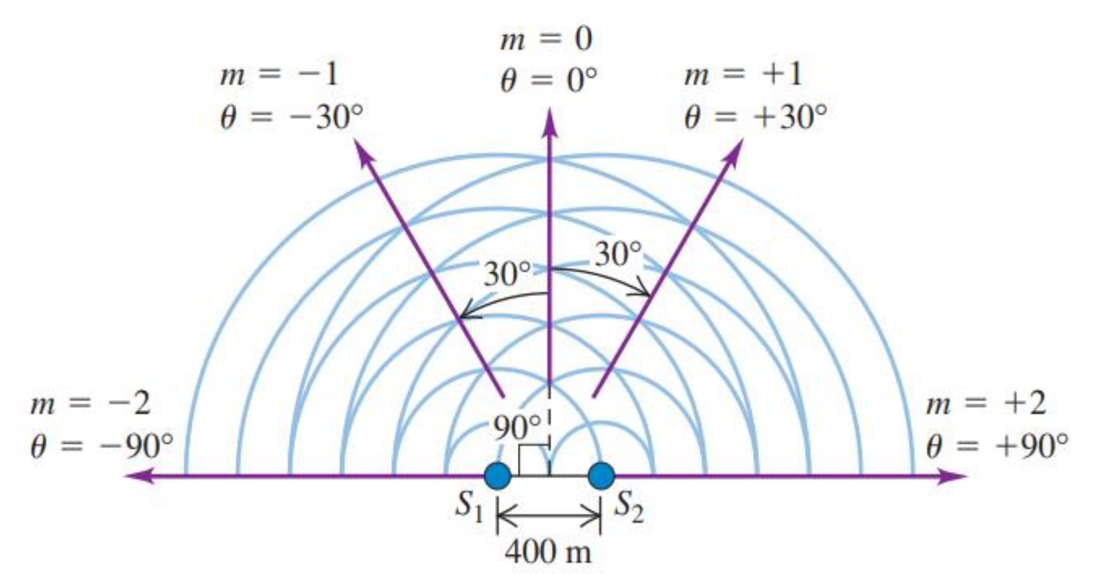


Solution:

$$\Delta(\text{path difference}) = (n-1)e = (2m+1)\frac{\lambda}{2}(m = 0, \pm1, \pm2, \cdots)$$

$$e_{\min} = \frac{\lambda/2}{n-1} = 533\text{nm}$$

It is often desirable to radiate most of the energy from a radio transmitter in particular directions rather than uniformly in all directions. Pairs or rows of antennas are often used to produce the desired radiation pattern. As an example, consider two identical vertical antennas apart, operating at 1500k Hz (near the top end of the AM broadcast band) and oscillating in phase. At distances much greater than in what directions is the intensity from the two antennas greatest?



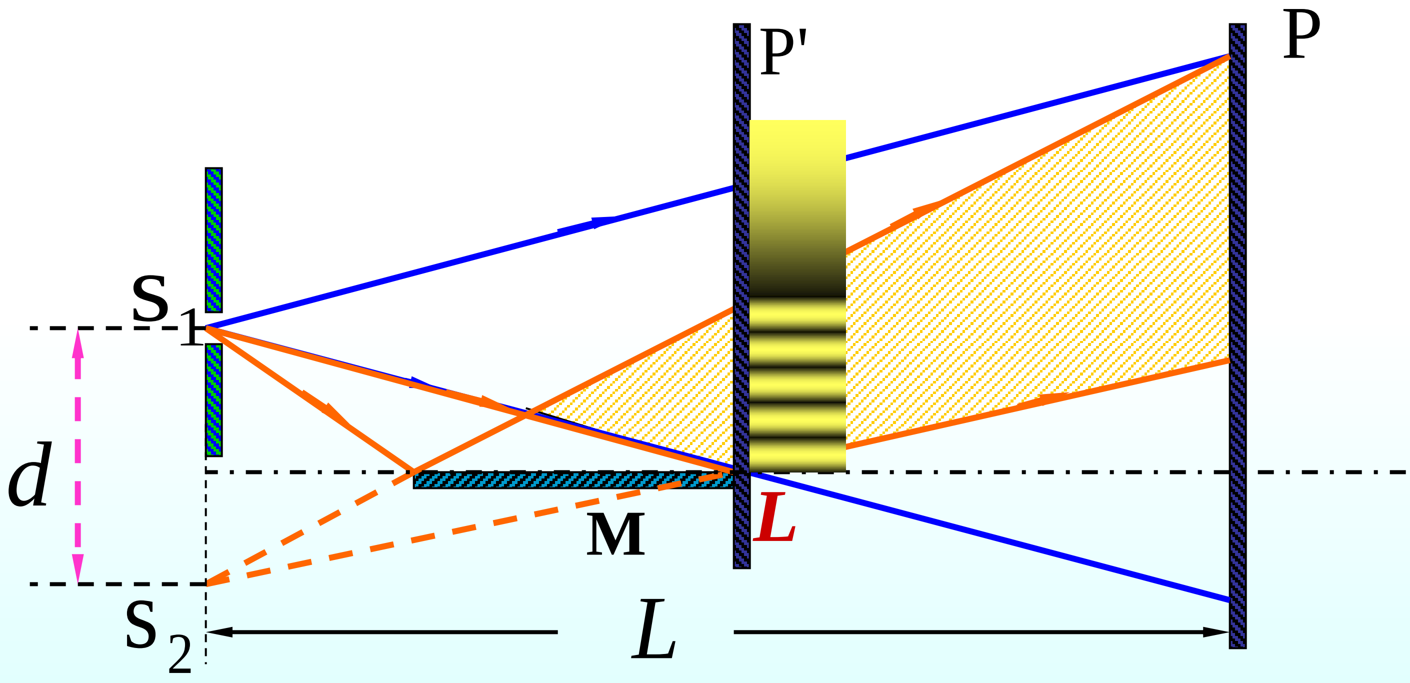
$$\lambda = \frac{c}{f} = 200\ m$$

$$d \sin\theta = m\lambda$$

$$\sin\theta = \frac{m\lambda}{d} = \frac{m}{2}$$

$$\theta = 0,\ 30^0,\ \ 90^0$$

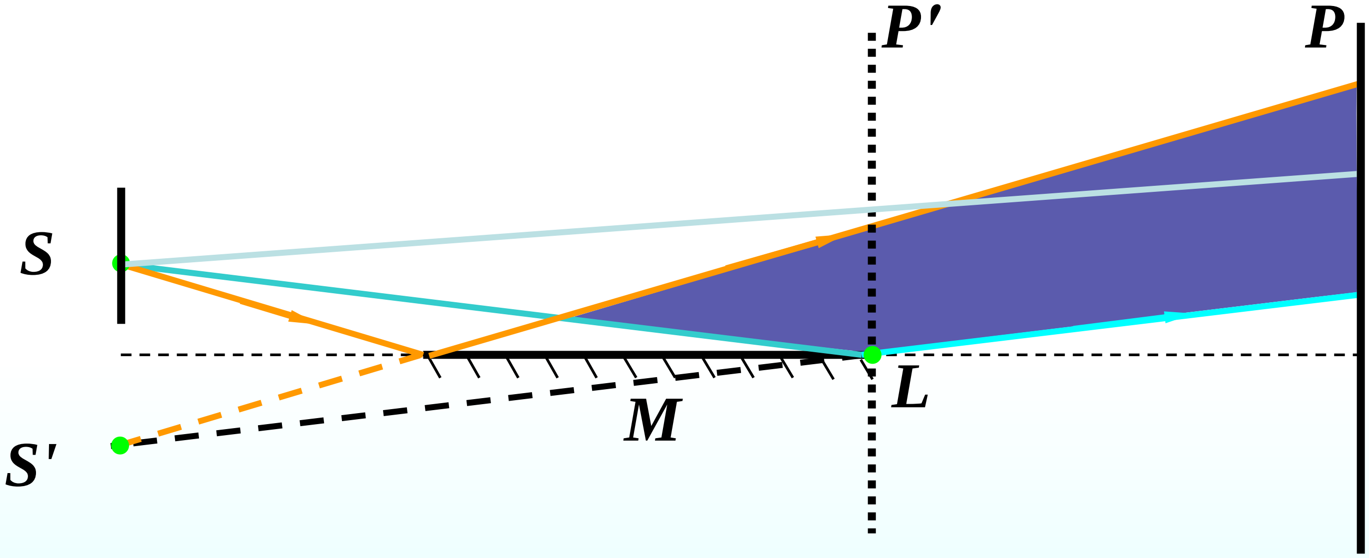
Lloyd mirror: provides one way of obtaining a double-slit interference pattern from a single source



Actual reflecting surface:

crystals for x-rays, ordinary glass for light,
wire screening for microwave,
a lake or even the earth’s ionosphere for radio wave

The point of contact of the mirror and the screen is dark → **half-wavelength loss**



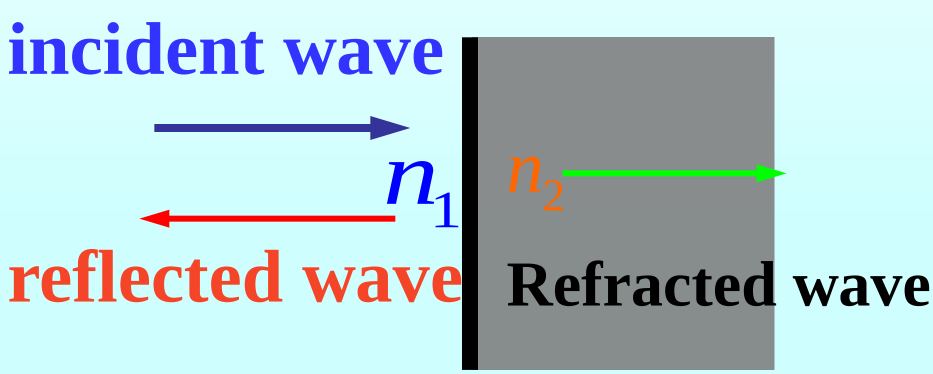
Change of phase due to reflection

$n_1 < n_2$

•**With half-wavelength loss**

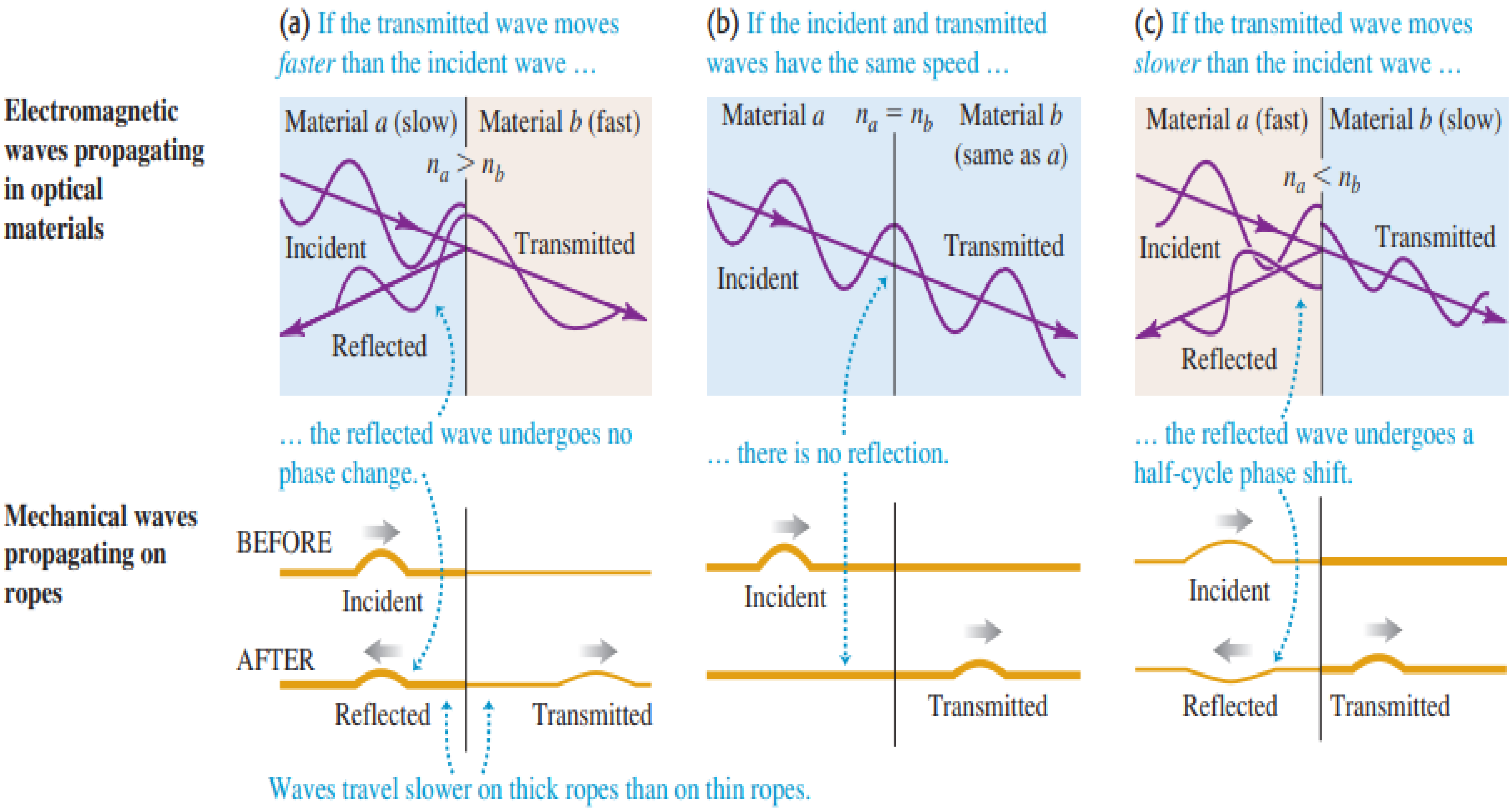
$n_1 > n_2$

•**No half-wavelength loss**

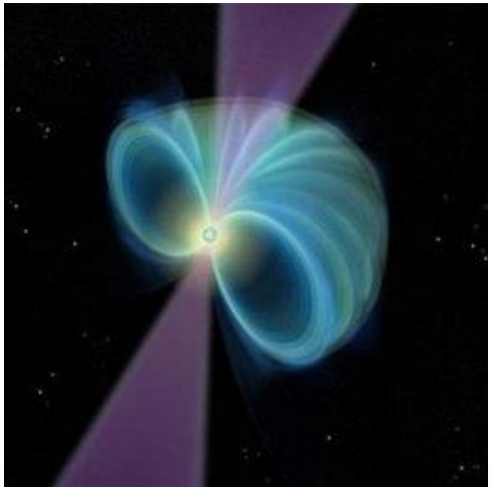


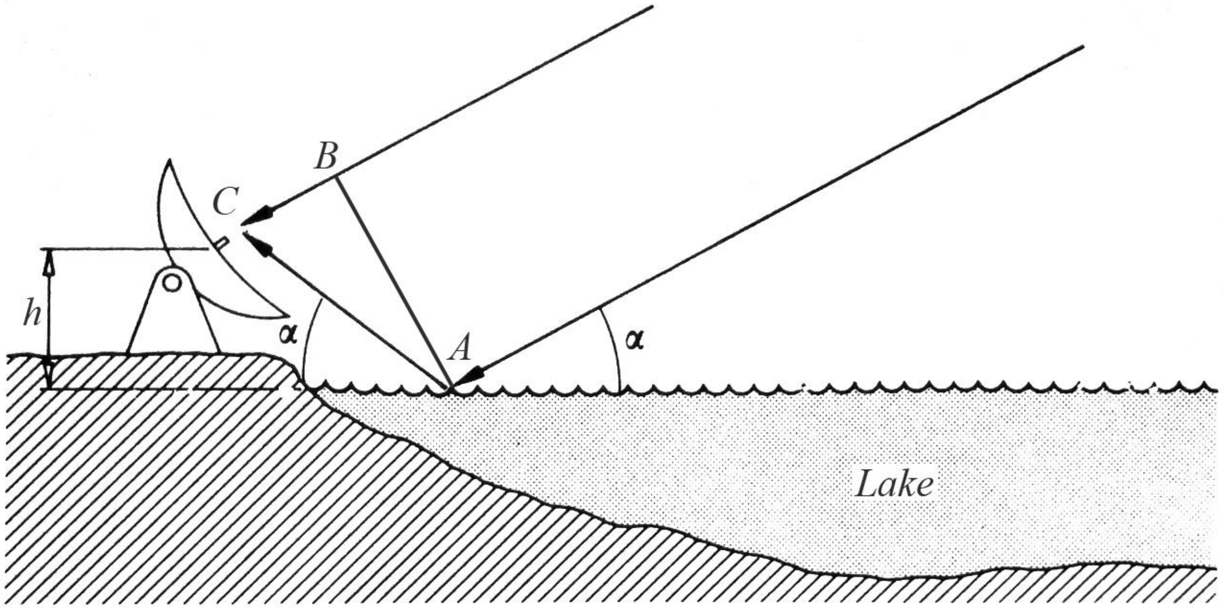
There is no half-wavelength loss in refracted waves.

Upper figures: electromagnetic waves striking an interface between optical materials at normal incidence (shown as a small angle for clarity). Lower figures: mechanical wave pulses on ropes.



5. **Radio star**





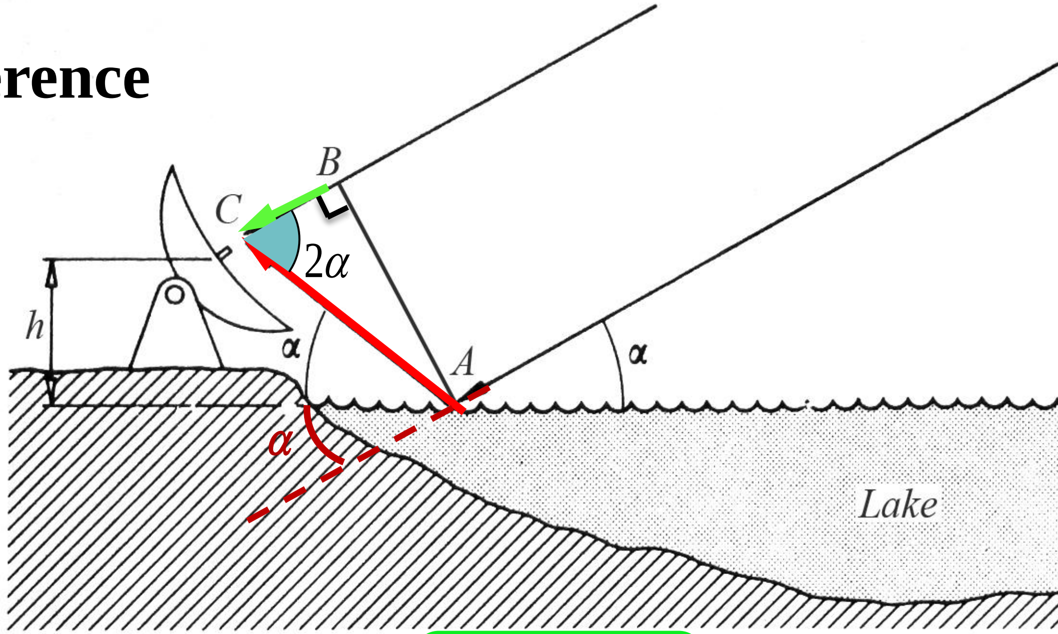
如图 离湖面 h=**0.5m** 处有一电磁波接收器位于 C，当一射电星从地平面渐渐升起时， 接收器断续地检测到一系列极大值．已知射电星所发射的电磁波的波长为 **20.0 cm**, 求第一次测到极大值时，射电星的方位与湖面所成的角度.

Solution: **Optical path difference**

$$\Delta = AC - BC + \frac{\lambda}{2}$$

$$= AC(1-\cos 2\alpha) + \frac{\lambda}{2}$$

$$= \frac{h}{\sin\alpha}(1-\cos 2\alpha) + \frac{\lambda}{2}$$



$2\sin^2\alpha$

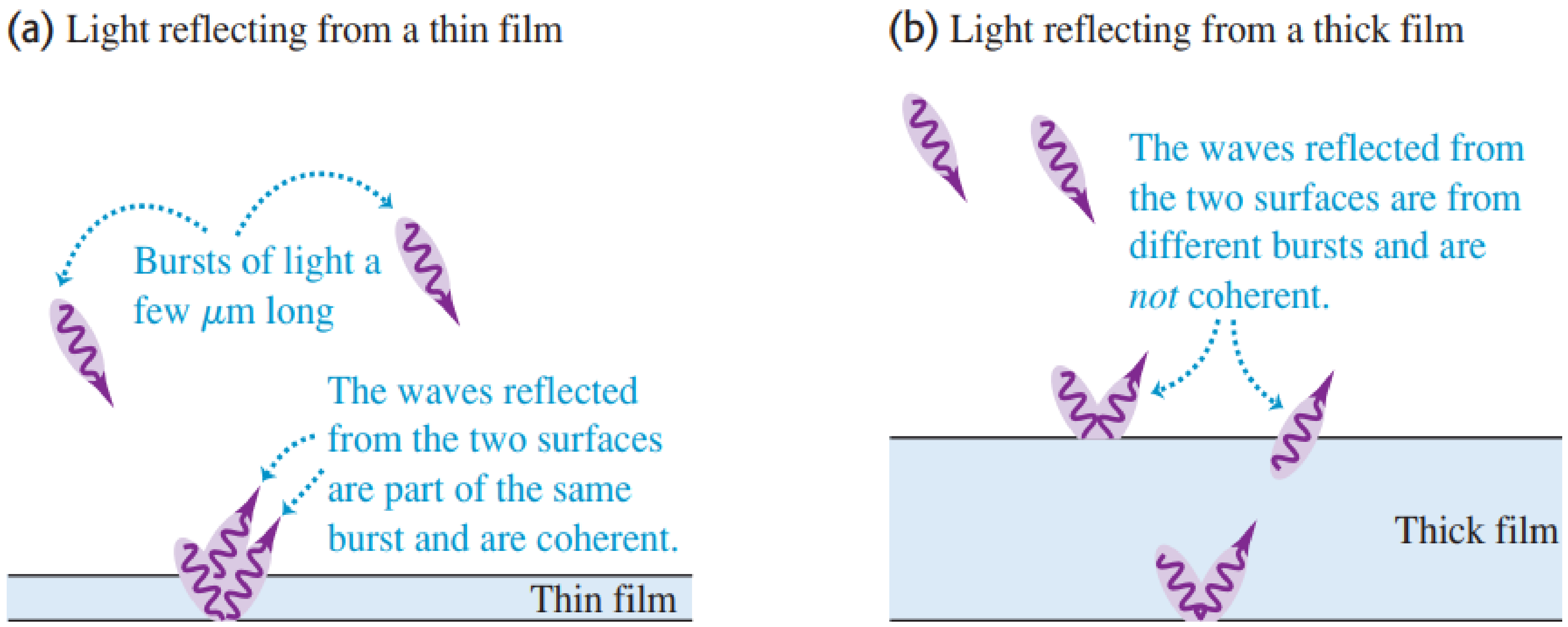
The maxima occur at $$\frac{h}{\sin\alpha}(1-\cos 2\alpha) + \frac{\lambda}{2} = m\lambda$$

AC > ***BC***, ***m*** ≠ **0;** ***m***= **1, 2, 3,** … $\sin\alpha = \dfrac{(2k-1)\lambda}{4h}$

$h = 0.5$ m
$\lambda = 20.0$ cm

m**=1, the first maximum is at** $$\alpha = \arcsin\frac{\lambda}{4h} = 5.74°$$

18.4 Interference in thin film

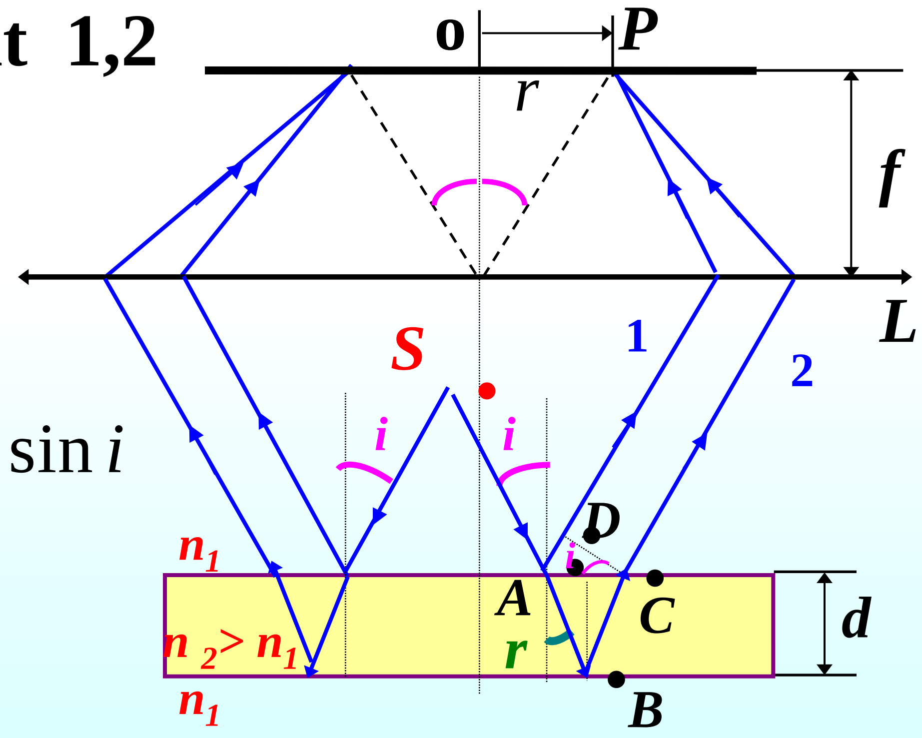


If the film is too thick, however, the two reflected waves will belong to different bursts. There is no definite phase relationship between different light bursts, so the two waves are incoherent and there is no fixed interference pattern.

18.4 Interference in thin film

1. Parallel plane thin film

optical path difference of light 1,2



$$\Delta = n_2(AB + BC) - n_1\ AD + \frac{\lambda}{2}$$

$$AB = BC = \frac{d}{\cos r}$$

$$AD = AC \cdot \sin i\ = 2d \cdot \tan r \cdot \sin i$$

$$\therefore \Delta = \frac{2n_2 d}{\cos r} - \frac{2n_1\ d \cdot \sin r \cdot \sin i}{\cos r} + \frac{\lambda}{2}$$

$$n_1 \sin i = n_2 \sin r$$

refraction law

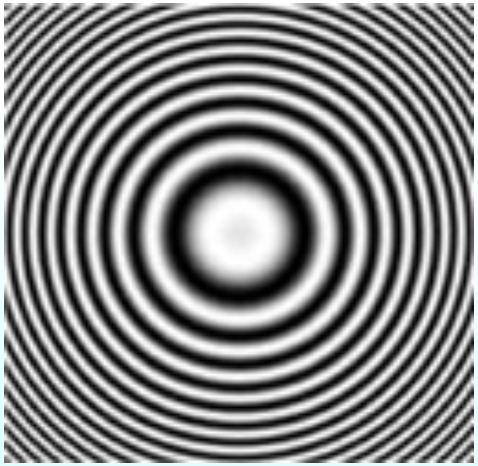
$$\Delta = 2n_2 d \cos r + \frac{\lambda}{2}$$ **or** $$\Delta = 2d\sqrt{n_2^{\ 2} - n_1^{\ 2}\sin^2 i} + \frac{\lambda}{2}$$

$$\Delta = 2d\sqrt{n_2^{\ 2} - n_1^{\ 2}\sin^2 i} + \frac{\lambda}{2} = 2n_2 d\cos r + \frac{\lambda}{2} = \Delta(i, d)$$

Discussion:

(1) Fixing *d* :equal inclination interference 等倾干涉

Because of the same thickness, Δ is only related to angle *i*. Therefore lights with the same *i* produce the one interference stripe.



Bright fringe $\Delta(i) = m\lambda, \quad m = 1, 2, 3, \cdots$

Dark fringe $\Delta(i) = (2m+1)\dfrac{\lambda}{2}, \quad m = 0, 1, 2, \cdots$

Dynamic changes of the rings:

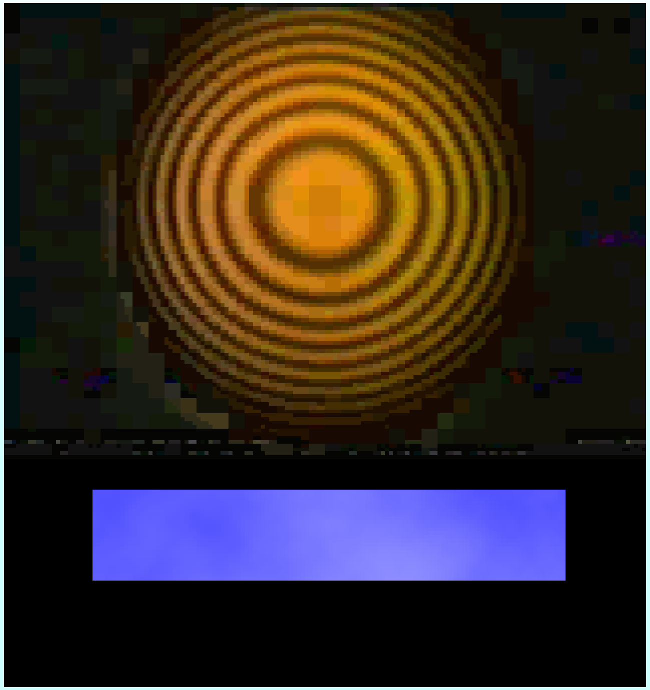
$$\Delta = 2n_2 d\cos r + \frac{\lambda}{2} = \Delta(i,d)$$

thickness of the film （*d*） changes → **The order of *m* changes**

Center of the rings presents light and dark alternative.

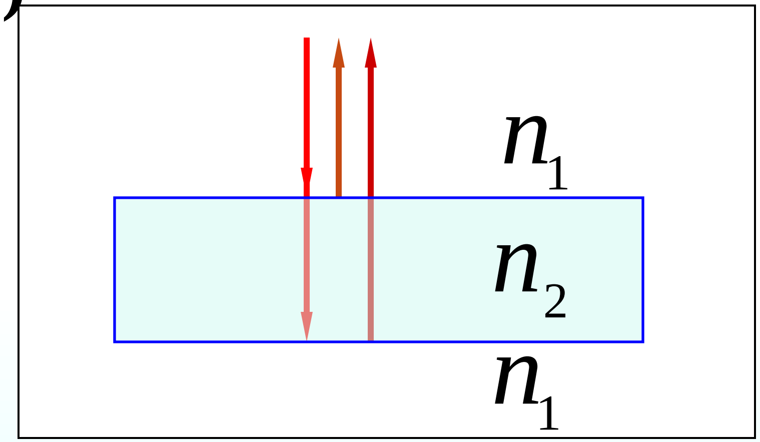
***d* reduced， rings “swallow”**

***d* increased， rings “threw up”**



(2) Fixing i : equal thickness interference 等厚干涉

For vertical incident(垂直入射)



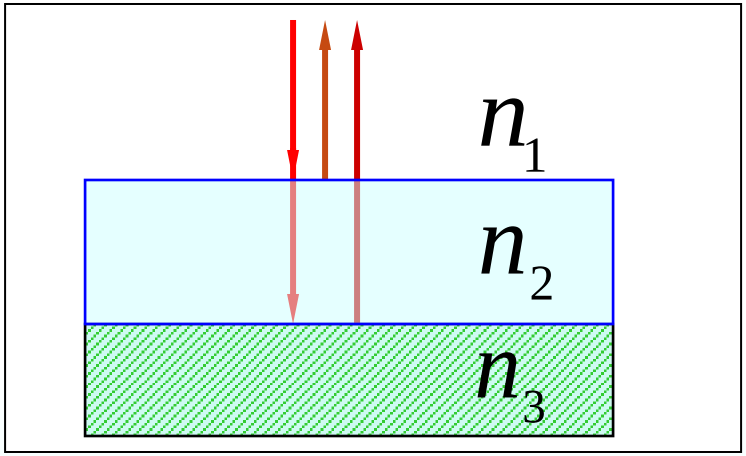
$$\Delta = 2n_2 d + \frac{\lambda}{2} = \Delta(d)$$

$$= \begin{cases} m\lambda & m = 1, 2, \cdots (bright) \\ (2m+1)\dfrac{\lambda}{2} & m = 0, 1, 2, \cdots (dark) \end{cases}$$

厚度***d***相同的光线对应同一条干涉条纹——等厚条纹

(3) half-wavelength loss $i = 0^0$

a. $n_1 < n_2 > n_3 \quad or \quad n_1 > n_2 < n_3$

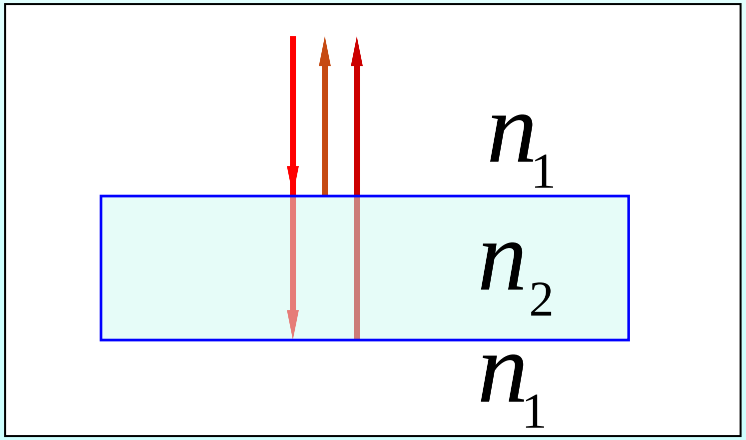


With half-wavelength loss

b. $n_1 > n_2 > n_3 \quad or \quad n_1 < n_2 < n_3$

No half-wavelength loss

c. $n_1 > n_2 < n_1 \quad or \quad n_1 < n_2 > n_1$

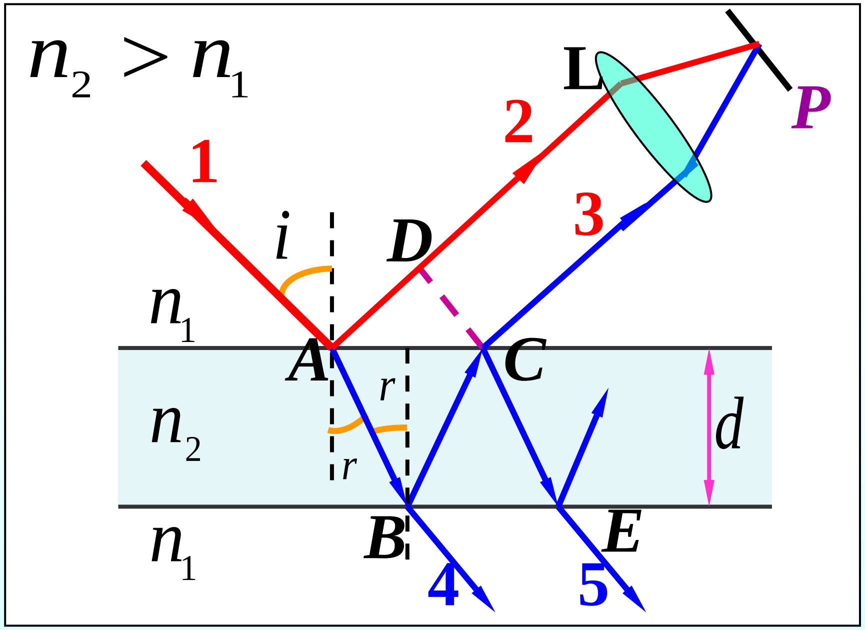


With half-wavelength loss

(4) transmission light 透射光

$$\varDelta_{\mathrm{r}} = 2d\sqrt{n_2^2 - n_1^2 \sin^2 i} + \boxed{\lambda / 2}$$

根据具体情况而定



➤ 透射光的光程差

$$\varDelta_{\mathrm{t}} = 2d\sqrt{n_2^2 - n_1^2 \sin^2 i}$$

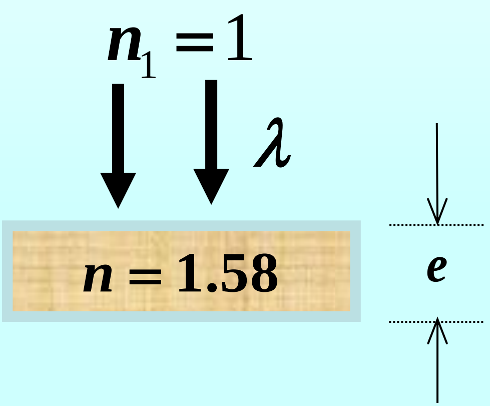
NOTE： refracted light and reflected light have complementarity, according with the law of conservation of energy.

6. **Monochromatic light of variable wavelength is incident normally on a thin sheet of plastic film in air. The reflected light is a minimum only for λ=510nm and λ=680nm in the visible spectrum. What is the thickness of the film (*n*=1.58).**

solution:

$$2nd+\frac{1}{2}\lambda_1=(m_1+\frac{1}{2})\lambda_1,\quad 2nd+\frac{1}{2}\lambda_2=(m_2+\frac{1}{2})\lambda_2$$

$$\frac{m_1}{m_2}=\frac{\lambda_2}{\lambda_1}=\frac{4}{3}\Rightarrow m_1=4,\quad m_2=3$$

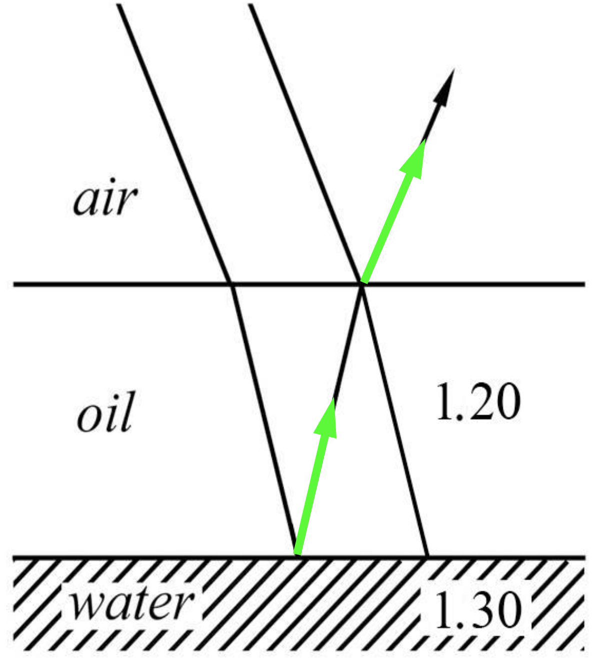


$$2nd=m_1\lambda_1\Rightarrow d=\frac{m_1\lambda_1}{2n}=\frac{4\times 510}{2\times 1.58}\text{nm}=646\text{nm}$$

7 **Known** $d = 460$ nm

What color would

(1) a pilot (2) a diver see?



Solution:(1)

$$\Delta = 2n_1 d = m\lambda$$

$$\lambda_1 = 2n_1 d = 2\times 1.20\times 460\,\text{nm} = 1.10\times 10^3\,\text{nm} \qquad m=1$$

$$\lambda_2 = n_1 d = 1.20\times 460\,\text{nm} = 552\,\text{nm} \qquad m=2 \quad \textbf{green}$$

$$\lambda_3 = \frac{2}{3} n_1 d = \frac{2}{3}\times 1.20\times 460\,\text{nm} = 368\,\text{nm} \qquad m=3$$

Only λ_2 is in the visible range-green.

(2) for transmission lights

$$\Delta = 2n_1 d + \frac{\lambda}{2}$$

Let

$$\Delta = m\lambda$$

We have $$\lambda_m = \frac{2n_1 d}{m - \frac{1}{2}}$$

Then

$\lambda_1 = 2.21 \times 10^3 \text{nm} \quad m = 1$

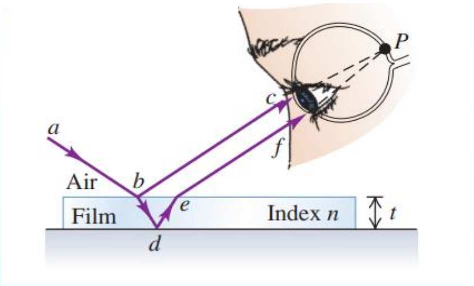
$\lambda_2 = 736\,\text{nm} \quad m = 2$ **red**

$\lambda_3 = 442\,\text{nm} \quad m = 3$ **violet**

$\lambda_4 = 315\,\text{nm} \quad m = 4$

Nonreflective coating **What is the minimum thickness of optical coating of MgF_2 which is designed to eliminate reflected light with** λ=550 nm **when incident normally on the glass.** n_1=1.00， n_2=1.38， n_3=1.50

Solution :

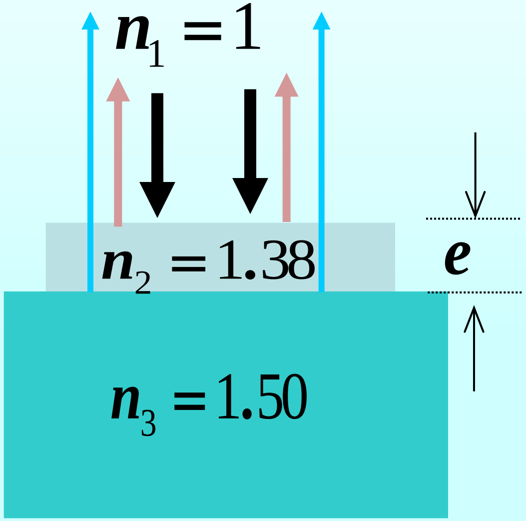


$$\because\ n_1 < n_2 < n_3$$

$$\varDelta = 2n_2 d$$

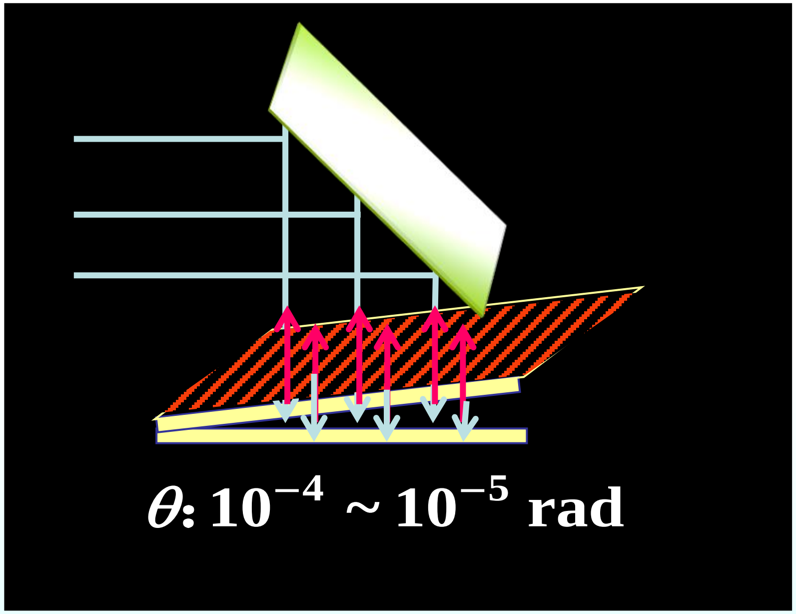
$$= (2m+1)\frac{\lambda}{2}$$

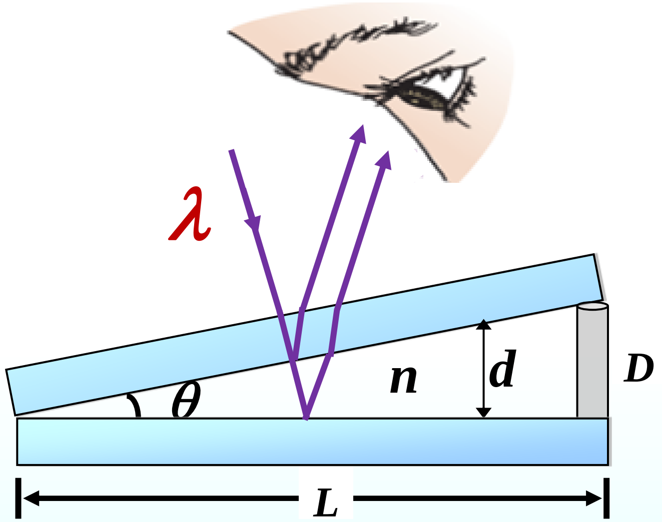
Transmission enhanced film



$$m = 0 \qquad e_{\min} = \frac{\lambda}{4n_2} = 99.6\ \text{nm}$$

2. Thin film of being wedge-shaped（等厚干涉）



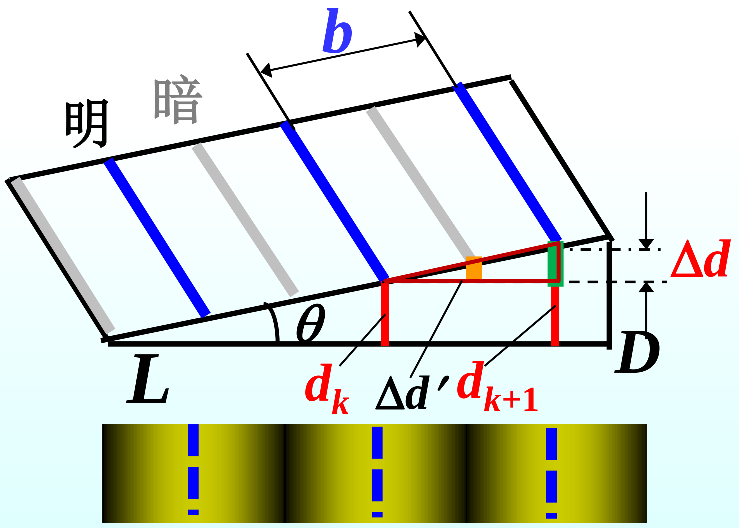


Pattern : a series straight fringes—fringe of equal thickness

$$\Delta = 2nd + \frac{\lambda}{2} = \begin{cases} m\lambda, \quad m = 1, 2, \cdots & \textbf{Bright fringe} \\ (2m+1)\dfrac{\lambda}{2}, \quad m = 0, 1, \cdots & \textbf{Dark fringe} \end{cases}$$

Discussion

$$\varDelta = 2nd + \boxed{\frac{\lambda}{2}}$$



$$\theta \approx \frac{\Delta d}{b} = \frac{\lambda}{2nb} = \frac{D}{L}$$

(1)棱边处 $d = 0$ 为暗纹.

(2) 相邻明纹（暗纹）

间的厚度差 $\Delta d = \dfrac{\lambda}{2n}$

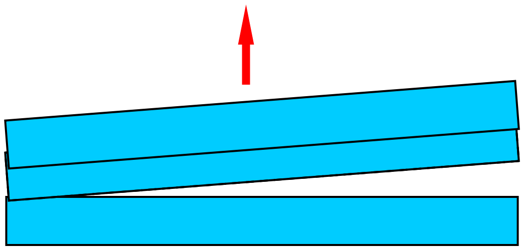
(3) 相邻明纹与暗纹

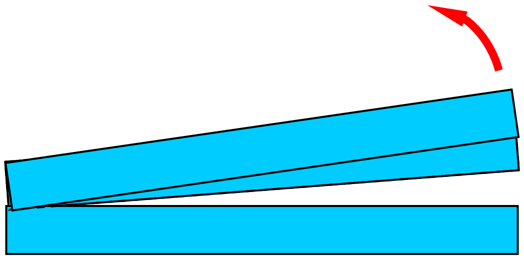
间的厚度差 $\Delta d' = \dfrac{\lambda}{4n}$

(4) 条纹间距

$$b = \frac{\Delta d}{\sin\theta} = \frac{\Delta d}{\theta} = \frac{\lambda}{2n\theta}$$

(5) 条纹的动态变化分析（n, λ, θ 变化时）

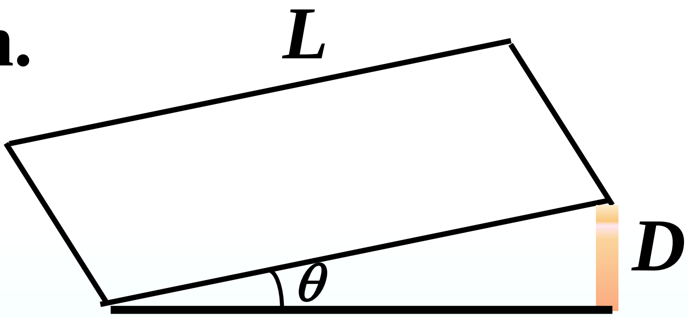




$$\Delta = 2nd + \frac{\lambda}{2} = m\lambda \qquad b = \frac{\Delta d}{\theta} = \frac{\lambda}{2n\theta} \qquad N = \frac{D}{\Delta d} = \frac{2nD}{\lambda}$$

- 上板向上平移：条纹向棱边移动，条纹间距不变，条纹数不变
- 劈尖角增大$\theta\uparrow$：条纹向棱边移动，条纹间距减小，条纹数增多
- 波长增大$\lambda\uparrow$：条纹向远离棱边移动，条纹间距增大，条纹数减少

Example A metal foil with D =0.048 mm separates one end of two pieces of flat glass. The 680-nm light is incident normally. How many **dark lines** will be observed on the glass surface with L=12 cm.



Solution:

$$\Delta = 2d + \frac{\lambda}{2}$$

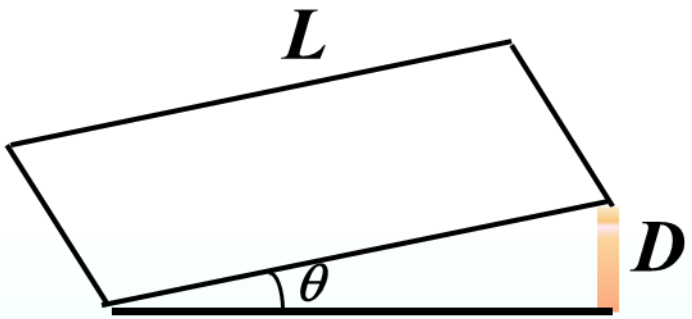
Destructive interference

$$2d + \frac{\lambda}{2} = (2m+1)\frac{\lambda}{2} \qquad m = 0, 1, 2, \cdots$$

$$2d_{\max} + \frac{\lambda}{2} = (2m_{\max}+1)\frac{\lambda}{2}$$

$$m_{\max} = \frac{2d_{\max}}{\lambda} = \frac{2D}{\lambda} = \frac{2\times 0.048\times 10^{-3}}{680\times 10^{-9}} = 141.1 \qquad N = \left[m_{\max}\right] + 1 = 142$$

A metal foil with D =0.048 mm separates one end of two pieces of flat glass. The 680-nm light is incident normally. How many dark lines will be observed on the glass surface with L=12 cm.



Solution 2:

相邻明纹（暗纹）间的厚度差

$$\Delta d = \frac{\lambda}{2n} = \frac{\lambda}{2} \qquad n=1$$

条纹间距数 $$N' = \frac{d_{\max}}{\Delta d} = \frac{D}{\Delta d} = \frac{2\times 0.048\times 10^{-3}}{680\times 10^{-9}} = 141.1$$

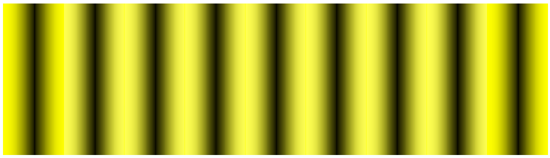
棱边为暗纹 $$N = N' + 1 = 142$$

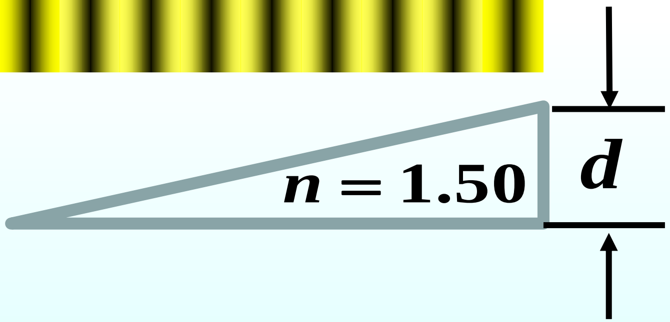
Example Light of wavelength 630nm is incident normally on a thin wedge-shaped film with index of refraction 1.50. There are ten bright and nine dark fringes over the length of the film. Find the thickness of the film.

Solution:

$$2nd = m\lambda$$

$$D = d_{\max} = \frac{9\lambda}{2\times 1.5} = 3\times 630nm = 1.89\mu m$$





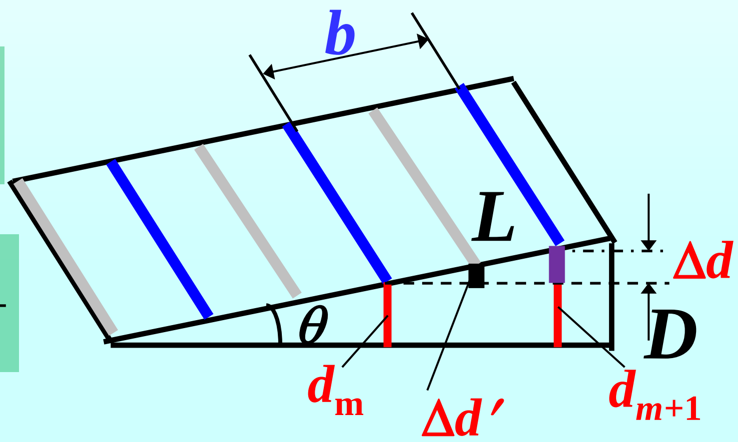
Solution 2:

Distance between two adjacent bright (dark) fringes

$$\Delta d = \frac{\lambda}{2n}$$

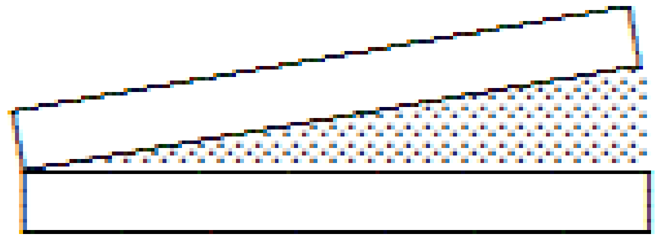
Distance between two adjacent bright fringe and dark fringe

$$\Delta d' = \frac{\lambda}{4n}$$



$$D = (9+10-1)\,\Delta d' = 1.89\mu m$$

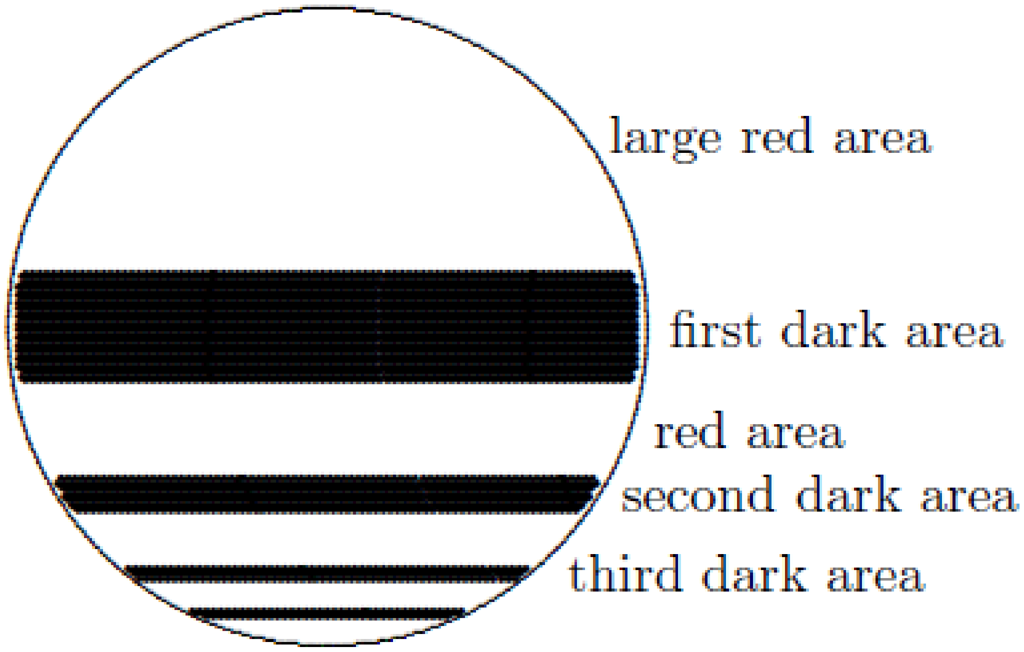
***Question*: A liquid of refractive index *n* = 4/3 replaces the air between a fixed wedge formed from two glass plates as shown. As a result, the spacing between adjacent dark bands in the interference pattern:**



A. increases by a factor of 4/3
B. increases by a factor of 3
C. remains the same
D. decreases to 3/4 of its original value
E. decreases to 1/3 of its original value

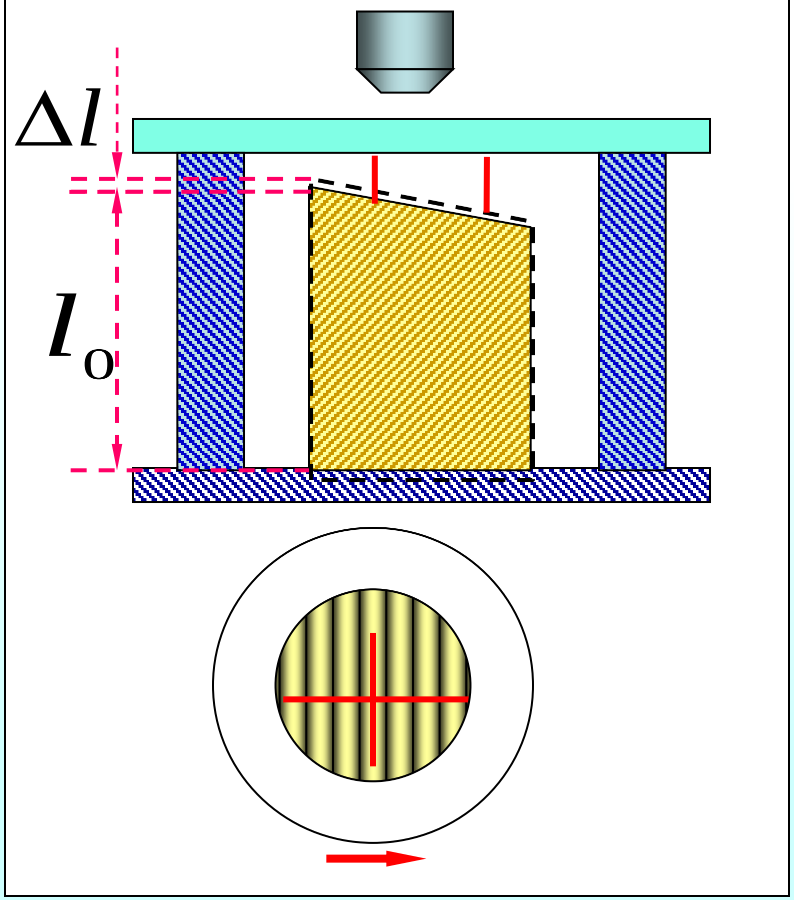
***Question*: Red light is viewed through a thin vertical soap film. At the third dark area shown, the thickness of the film, in terms of the wavelength within the film, is:**

A. λ
B. 3λ/4
C. λ/2
D. λ/4
E. 5λ/4



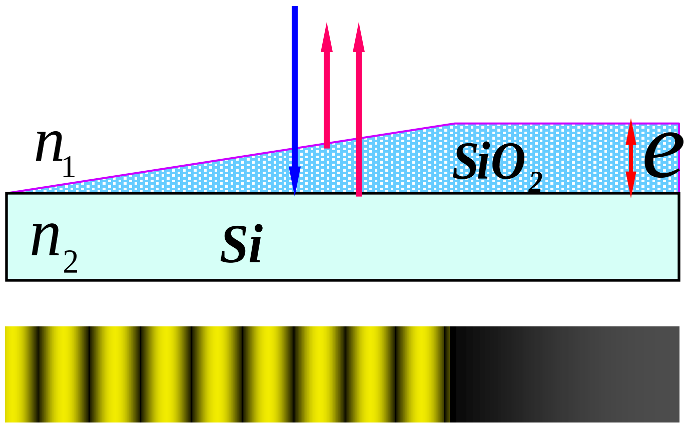
Application of interference of wedge-shaped thin film

(1) 干涉膨胀仪



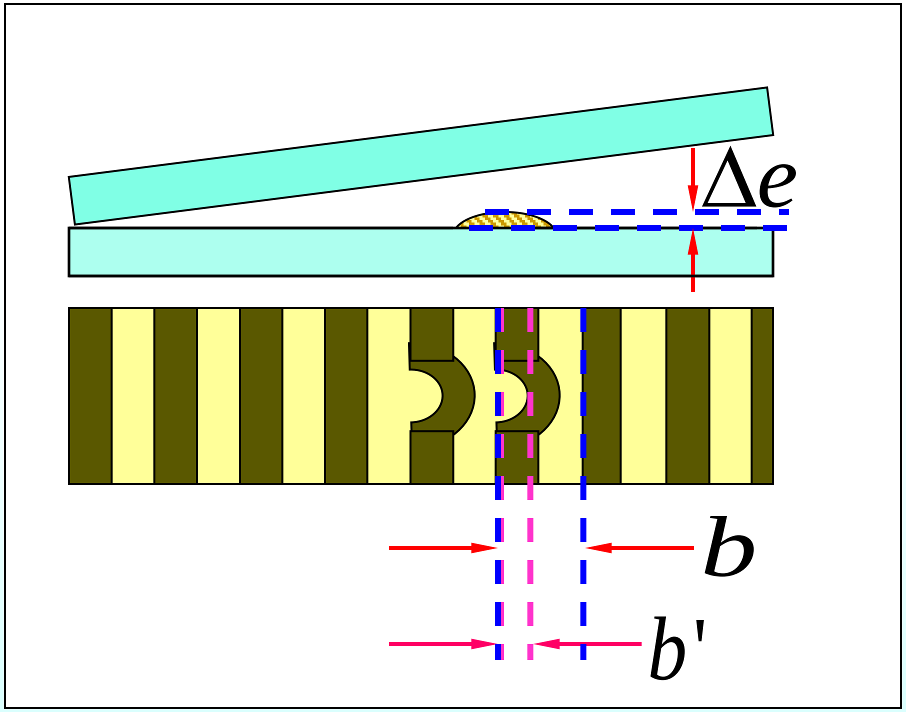
$$\Delta l = N\frac{\lambda}{2}$$

(2) 测膜厚



$$e = N\frac{\lambda}{2n_1}$$

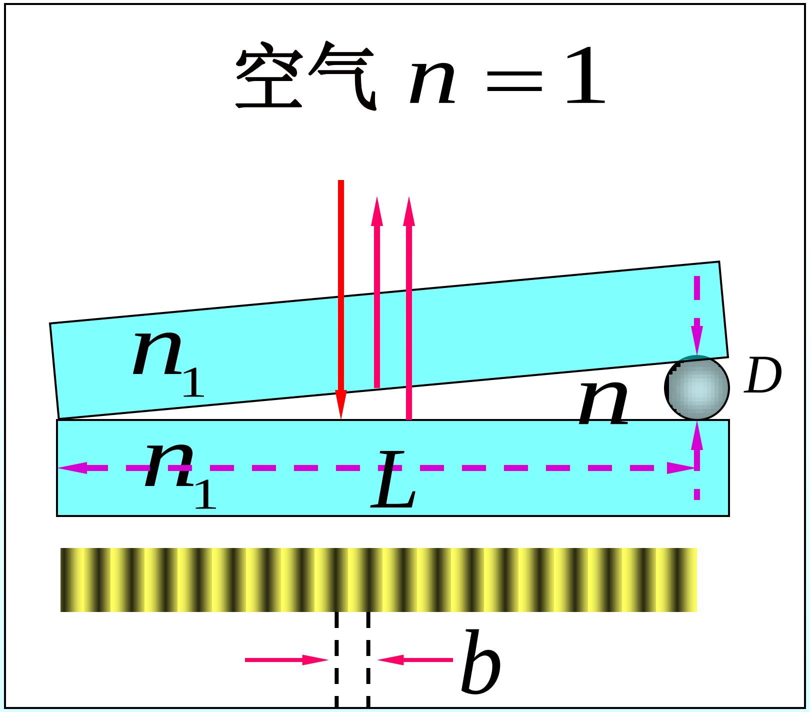
(3) 检验光学元件表面的平整度



$$\Delta e = \frac{b'}{b}\frac{\lambda}{2}$$

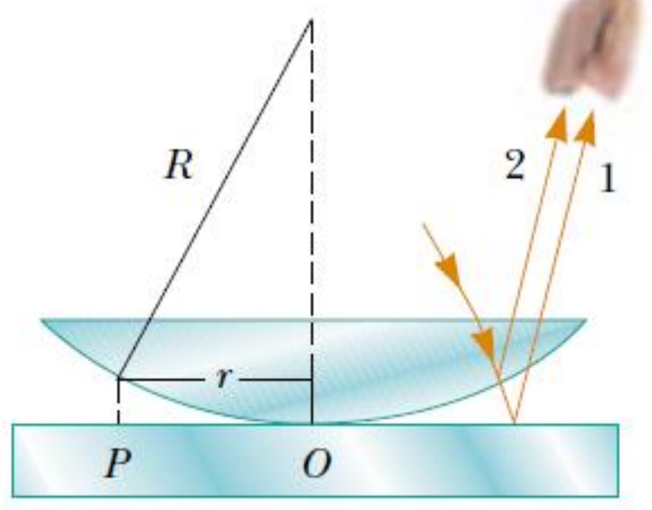
$$\approx \frac{1}{3}\cdot\frac{\lambda}{2} = \frac{\lambda}{6}$$

(4) 测细丝的直径

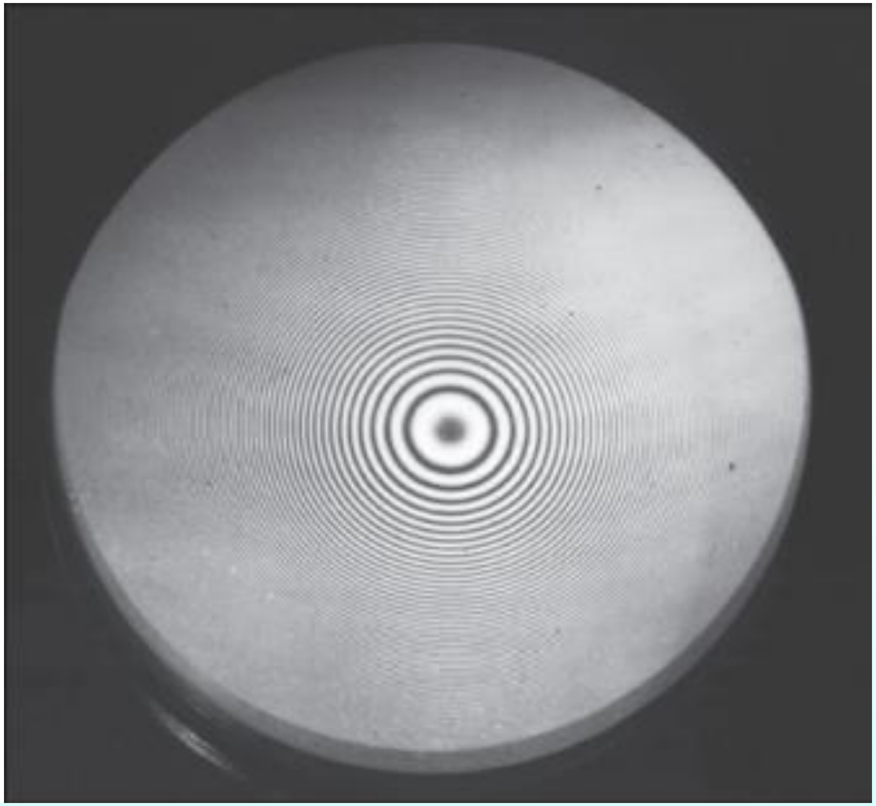


$$D = \frac{\lambda}{2n} \cdot \frac{L}{b}$$

3. Newton’s rings

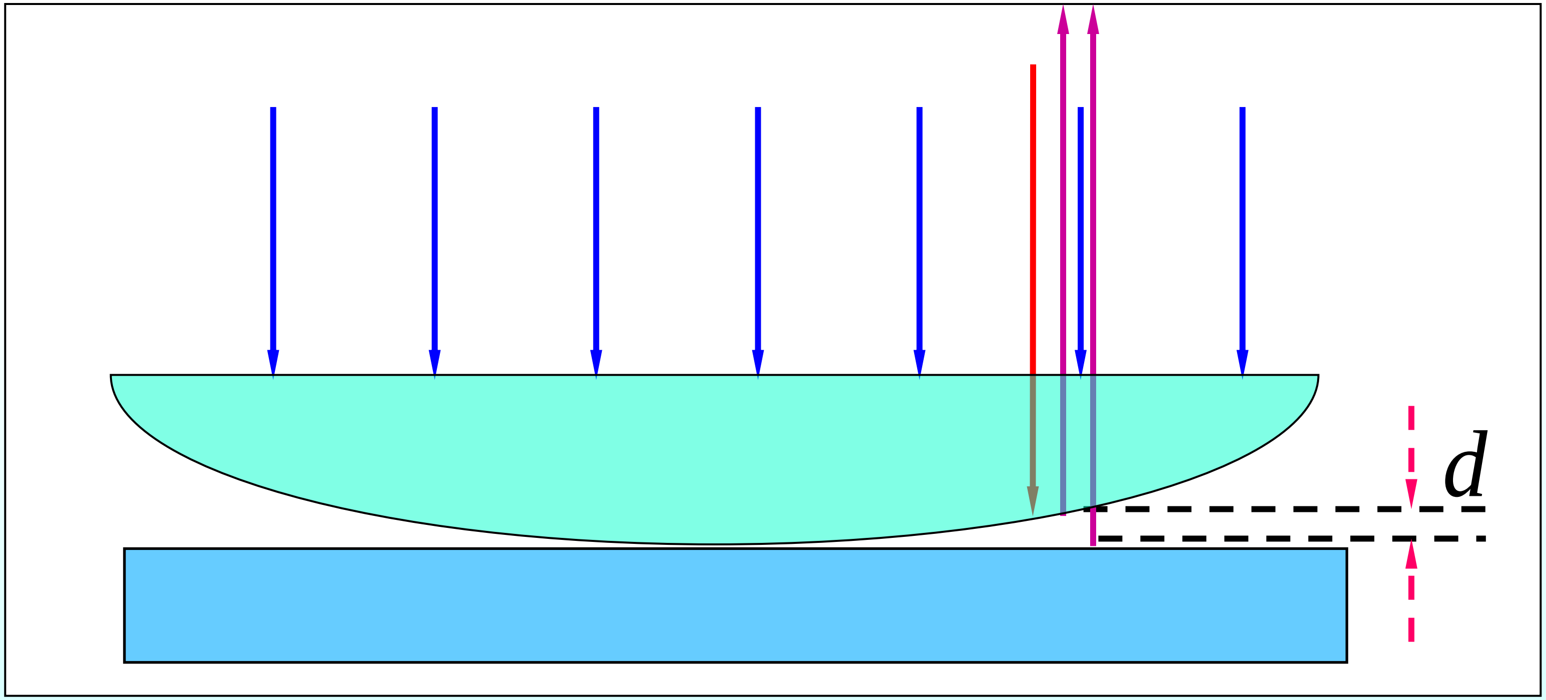


(a)



Rays reflected from the flat plate and the curved lens surface gives rise to an interference pattern known as Newton’s rings.

Consist of a flat glass and a planoconvex(平凸透镜）

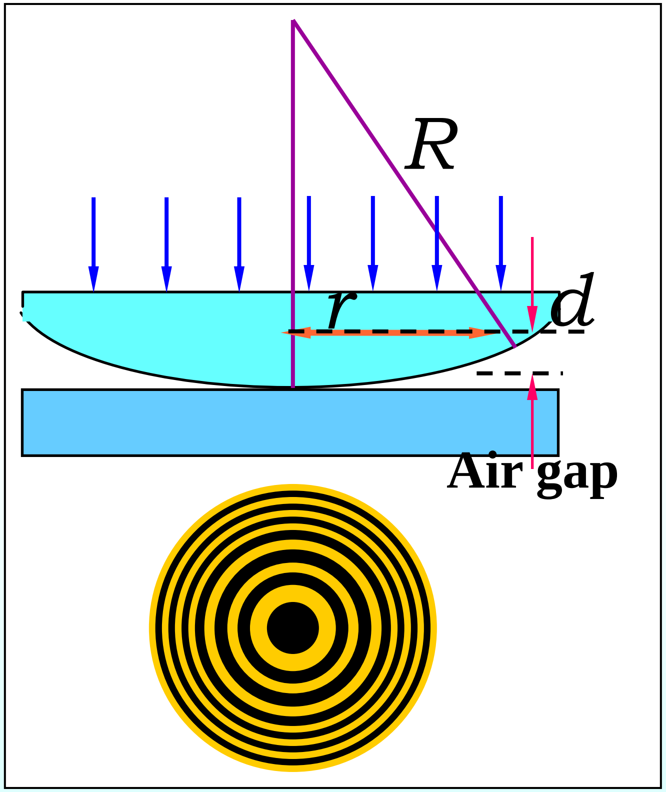


Optical Path difference

$$\Delta = 2d + \frac{\lambda}{2}$$

$$\Delta = 2d + \frac{\lambda}{2}$$

$$\Delta = \begin{cases} m\lambda \quad (m = 1, 2, \cdots) & \textbf{Bright fringe} \\ (2m+1)\dfrac{\lambda}{2} \quad (m = 0, 1, \cdots) & \textbf{Dark fringe} \end{cases}$$

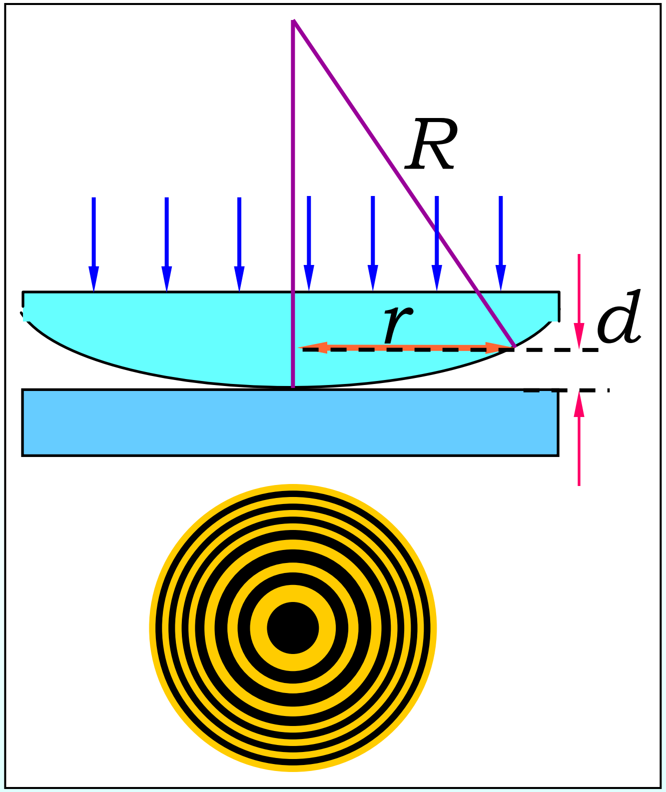


$$r^2 = R^2 - (R-d)^2 = 2dR - d^2$$

$$\because R >> d \quad \therefore d^2 \approx 0$$

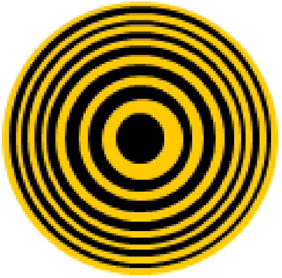
$$r^2 = 2dR = (\Delta - \frac{\lambda}{2})R$$

$$r = \sqrt{2dR} = \sqrt{(\Delta - \frac{\lambda}{2})R}$$



$$\Rightarrow \begin{cases} r_{mB} = \sqrt{(m - \frac{1}{2})R\lambda} & \textbf{Bright ring} \\ r_{mD} = \sqrt{mR\lambda} & \textbf{Dark ring} \end{cases}$$

Discussion: $r_D = \sqrt{mR\lambda}$



（1）无论入射光波长如何，中心永远是暗点。

（2）各亮纹彼此不等间距，半径越大，级次越高，相距越密集（里疏外密）

(3) 平凸透镜抬高（平动），条纹内陷，
平凸透镜压低（平动），条纹外涌。

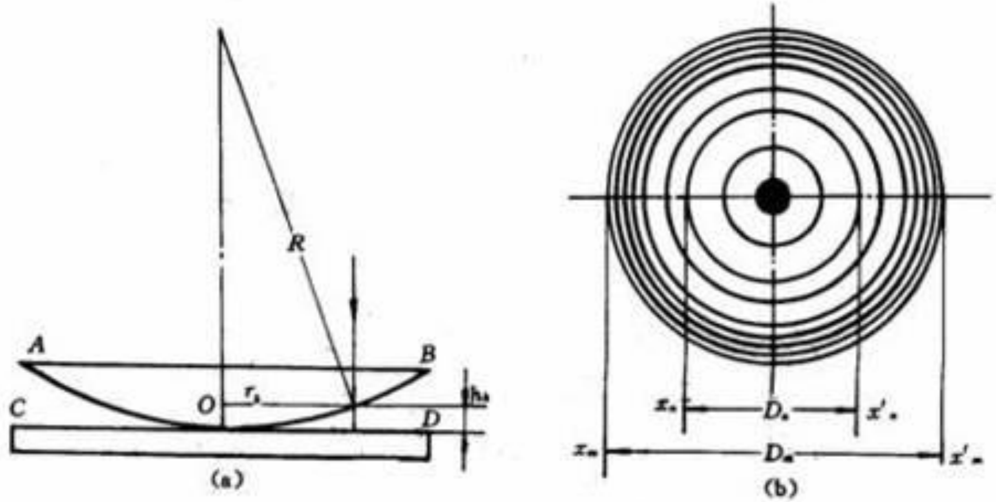
The applicaion of Newton's Rings

◆测透镜球面的半径 *R*

$$r_{mD}^{2} = mR\lambda$$

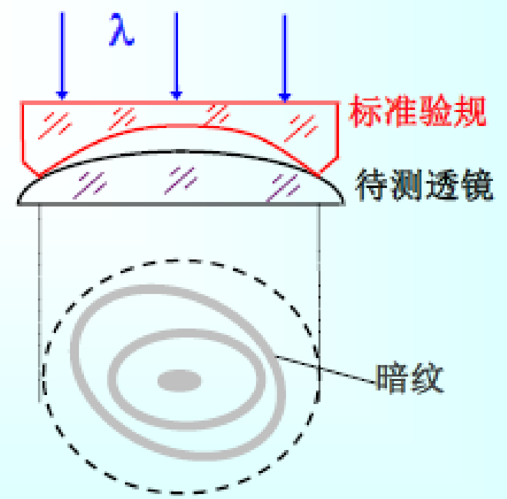
$$r_{(m+k)D}^{2} = (m+k)R\lambda$$

$$R = \frac{r_{m+k}^{2} - r_{m}^{2}}{k\lambda}$$

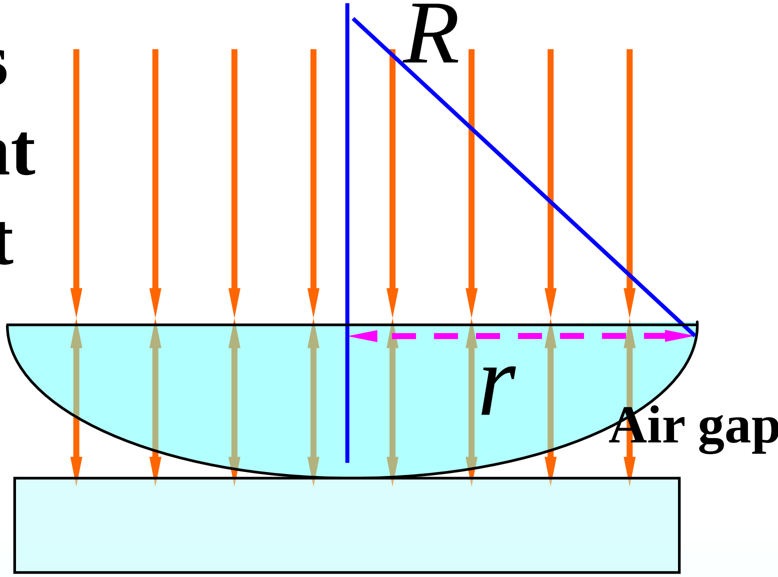


◆测量波长

◆检验透镜球面质量



Example **A planoconvex lucite lens 3.4cm in diameter is place on a flat piece of glass. When 580-nm light is incident normally, 48 bright rings are observed, the last one right at the edge. What is the radius of curvature of the lens surface?**



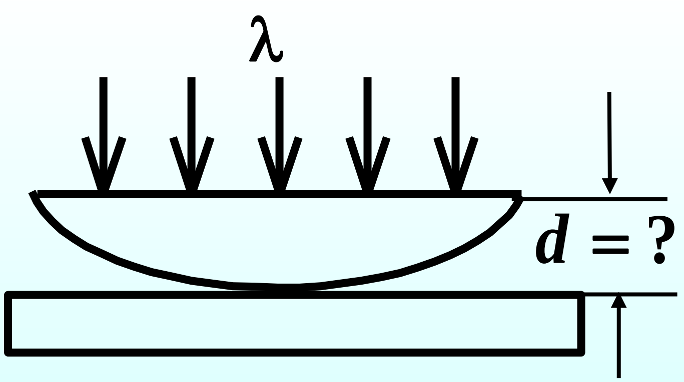
Solution: $$r_{mB} = \sqrt{(m - \frac{1}{2})R\lambda} \qquad m = 1, 2, \cdots$$

$$R = \frac{r^2{}_m}{(m - \frac{1}{2})\lambda} \qquad m = 48 \qquad R = 10.49m$$

Example: **A total of 28 bright and 28 dark Newton's rings (not counting the dark spot at the center) are observed when 650-nm light falls normally on a planoconvex lens resting on a flat glass surface. How much thicker is the center than the edges?**

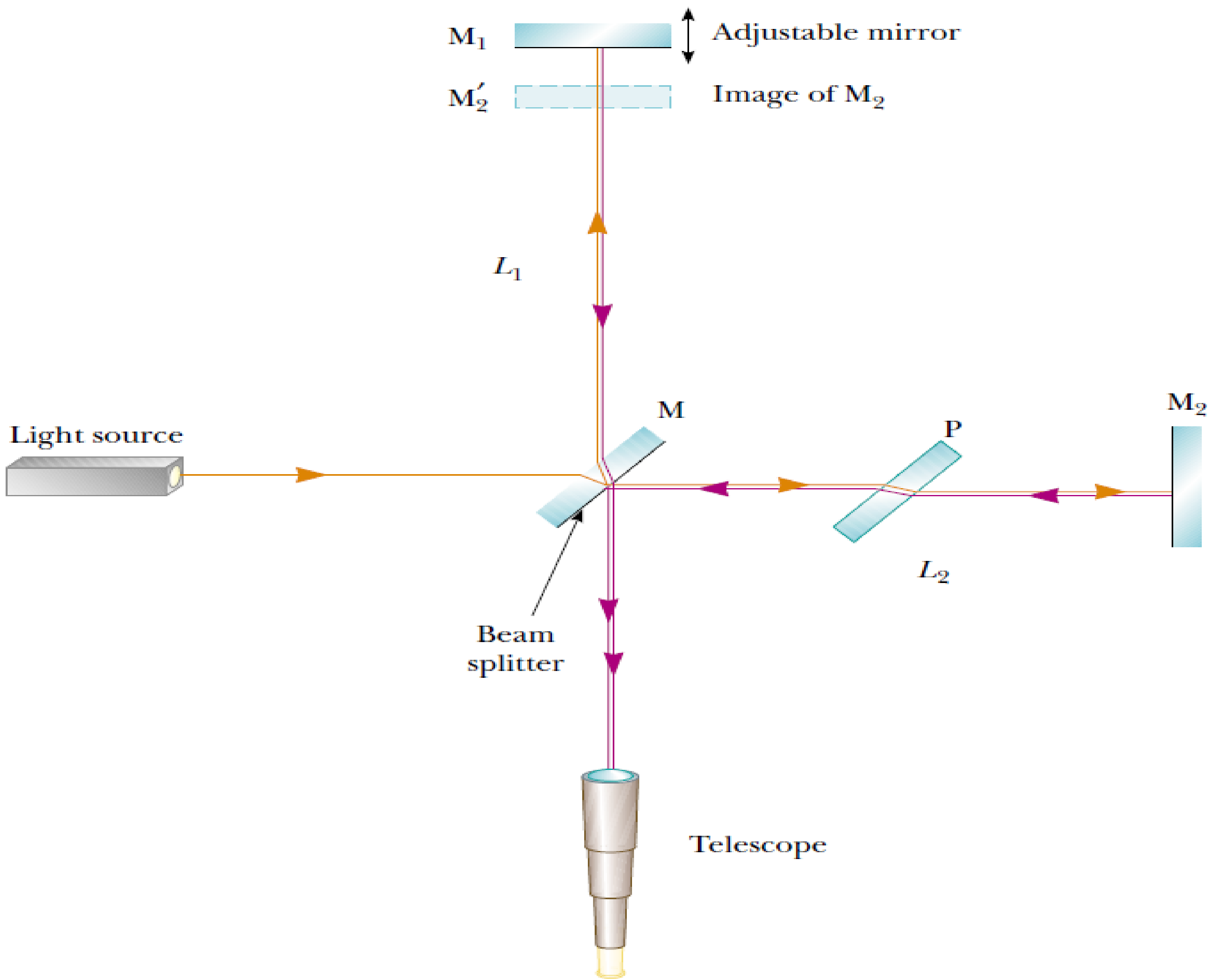
Solution:

$$2d + \frac{\lambda}{2} = (2m+1)\frac{\lambda}{2}$$



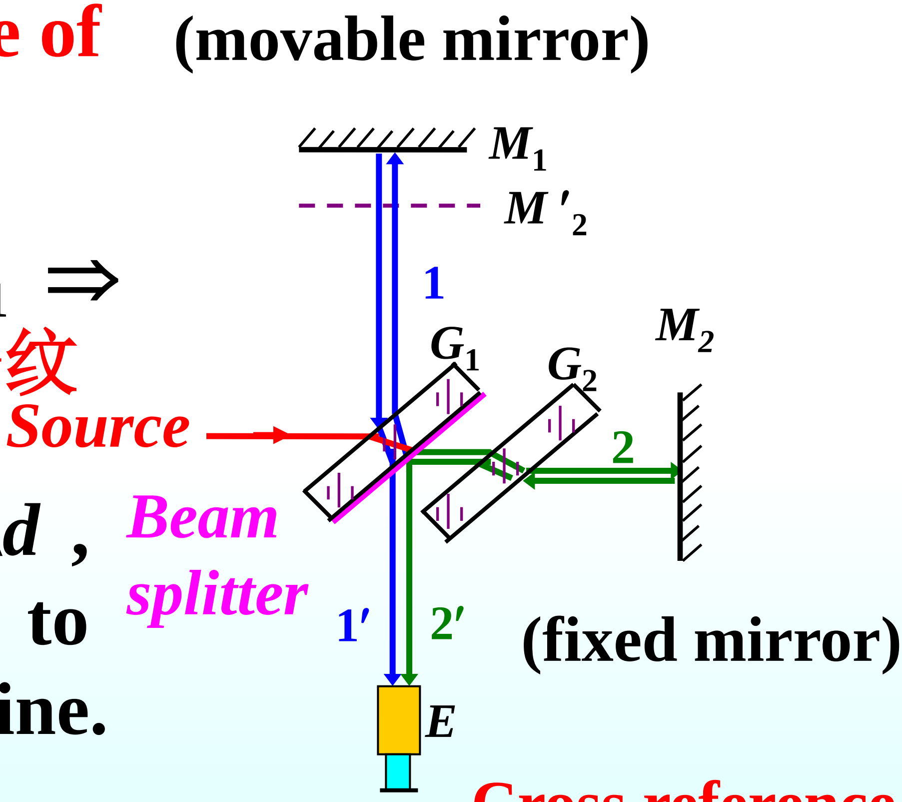
$$d = \frac{m\lambda}{2} = \frac{28 \times 650nm}{2} = 9.13\mu m$$

18.5 Michelson interferometer



•M'_2、M_1 are parallel $\Rightarrow$ **fringe of equal inclination 等倾条纹**

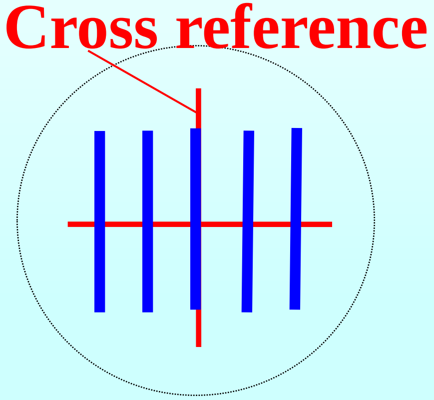
•A tiny angle between M'_2、M_1 $\Rightarrow$ **fringe of equal thickness等厚条纹**



If M_1 is moved the distance Δd, Δm fringes will be counted to move pass the cross reference line.

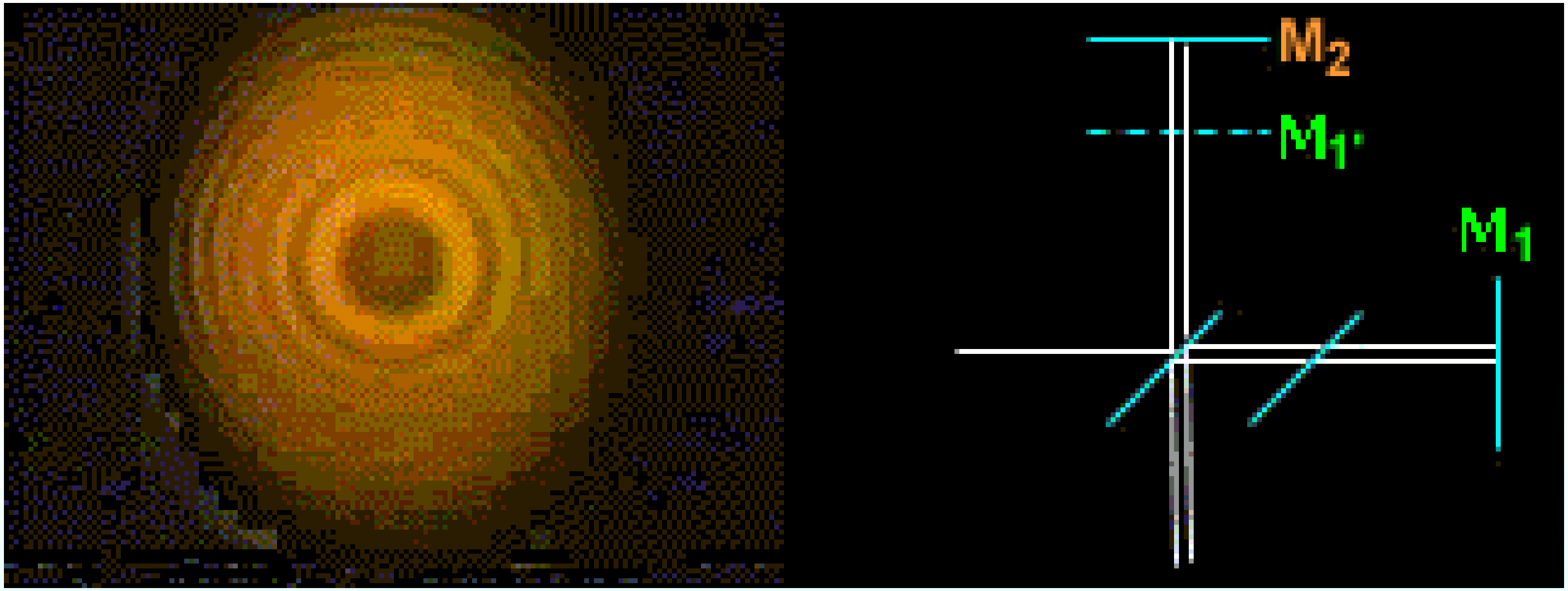
$$\boxed{2\Delta d = \Delta m \cdot \lambda}$$

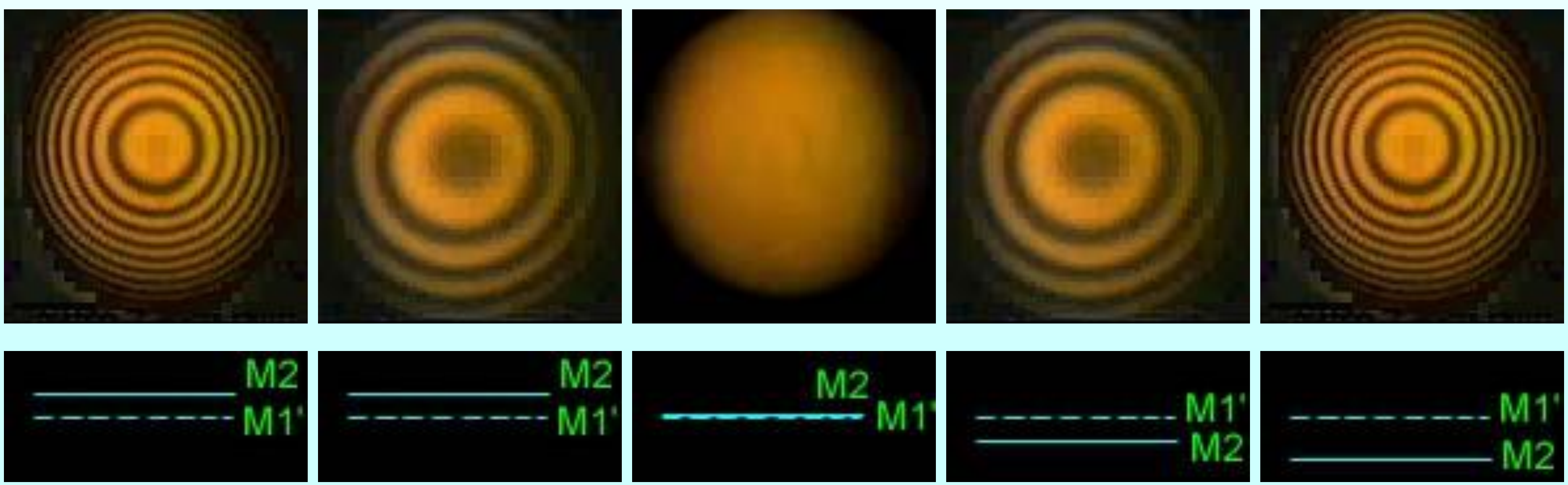
$$\boxed{\Delta d = \Delta m \cdot \frac{\lambda}{2}}$$



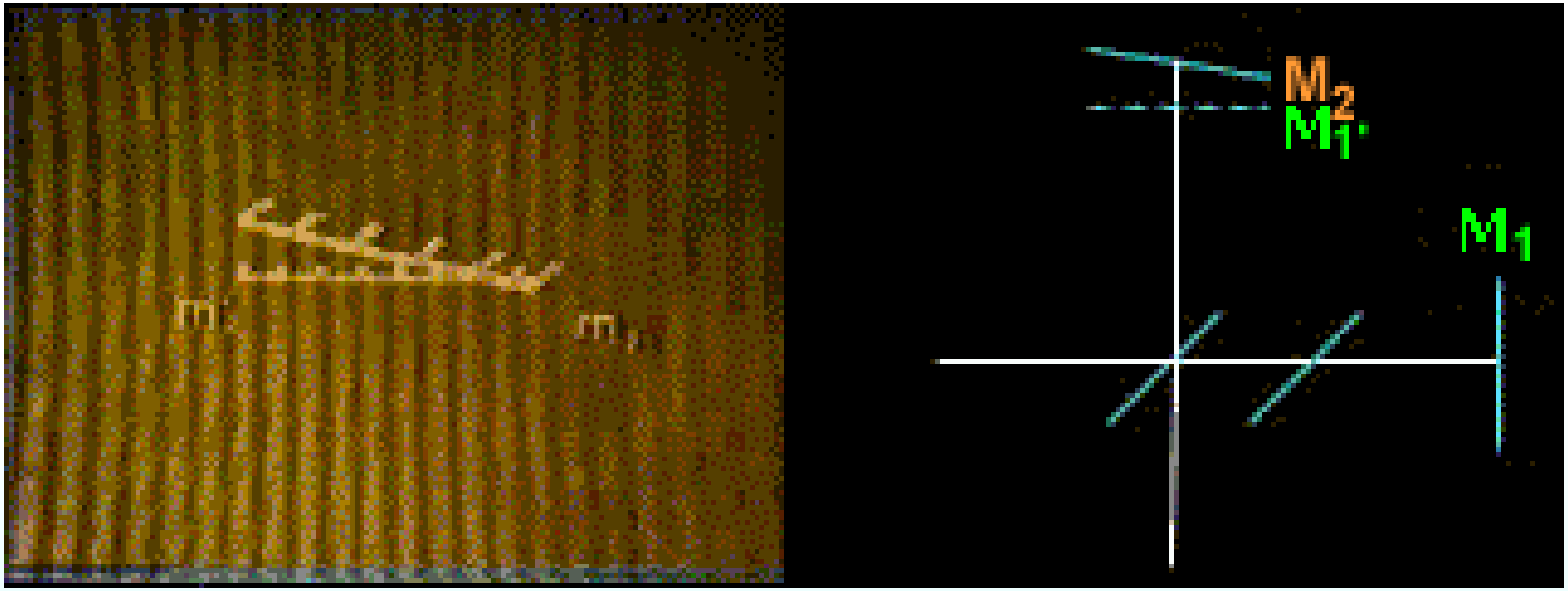
等厚条纹

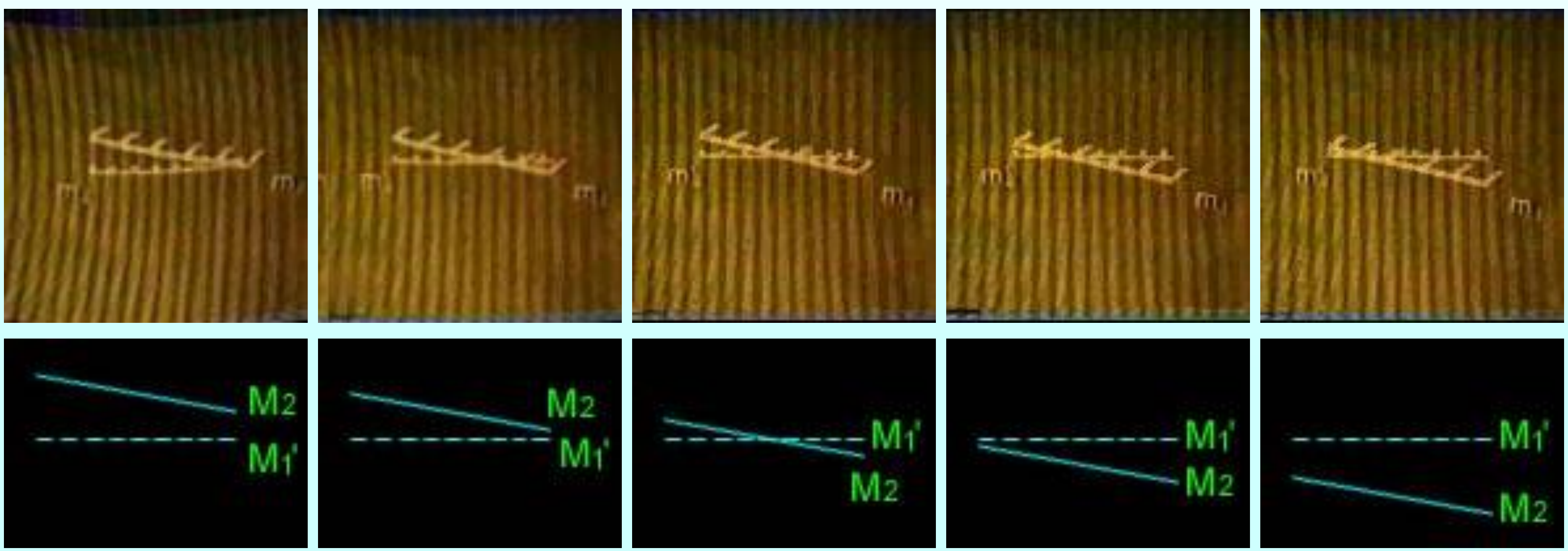
Experimental phenomena of parallel plane thin film



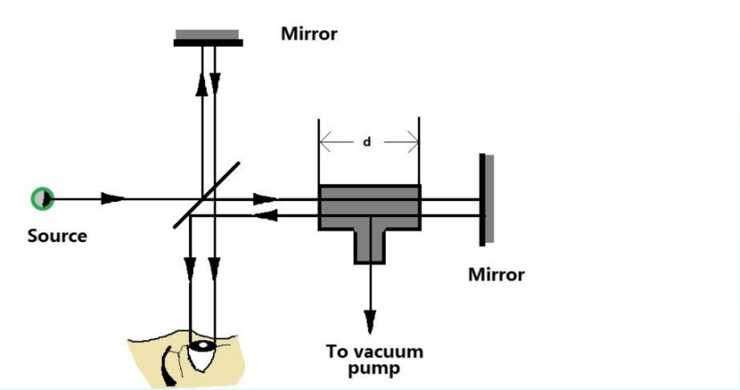


Experimental phenomena of equal thickness interference





Example One of the beam of the interferometer passes through a glass container of length 1.30cm, when a gas is allowed to slowly fill in a total 186 dark fringes are counted to move pass a reference line, the wavelength of light used is 610nm. Calculate the index of refraction of the gas.

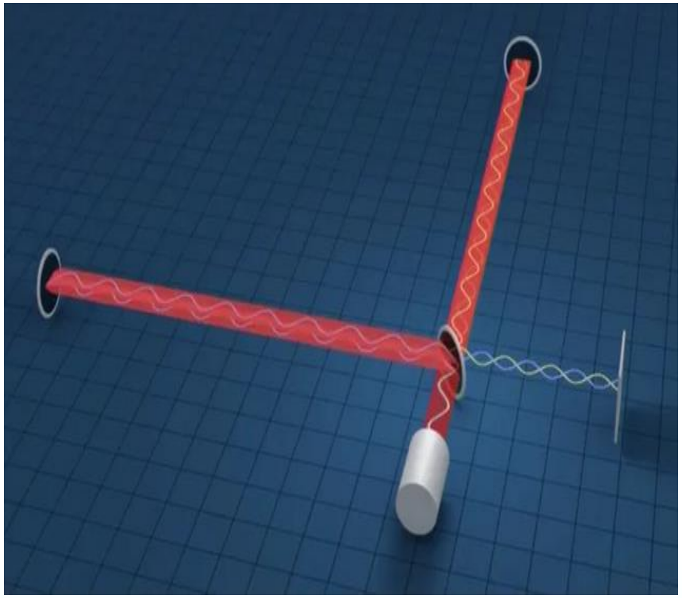


Solution:

$$2(n-1)l = 186\lambda$$

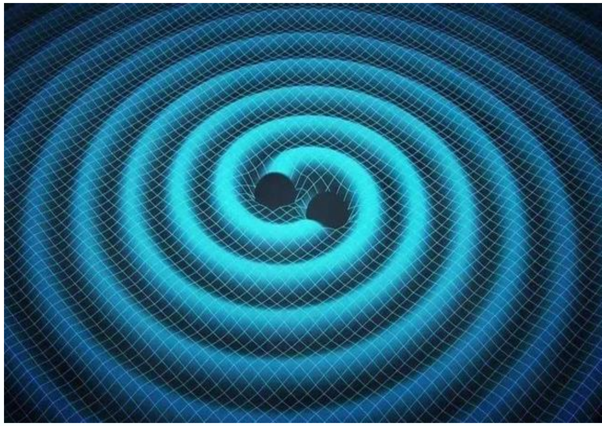
$$n = 1 + \frac{107.2\lambda}{2l} = 1 + \frac{186 \times 610 \times 10^{-9}}{2 \times 1.30 \ \times 10^{-2}} = 1.0044$$

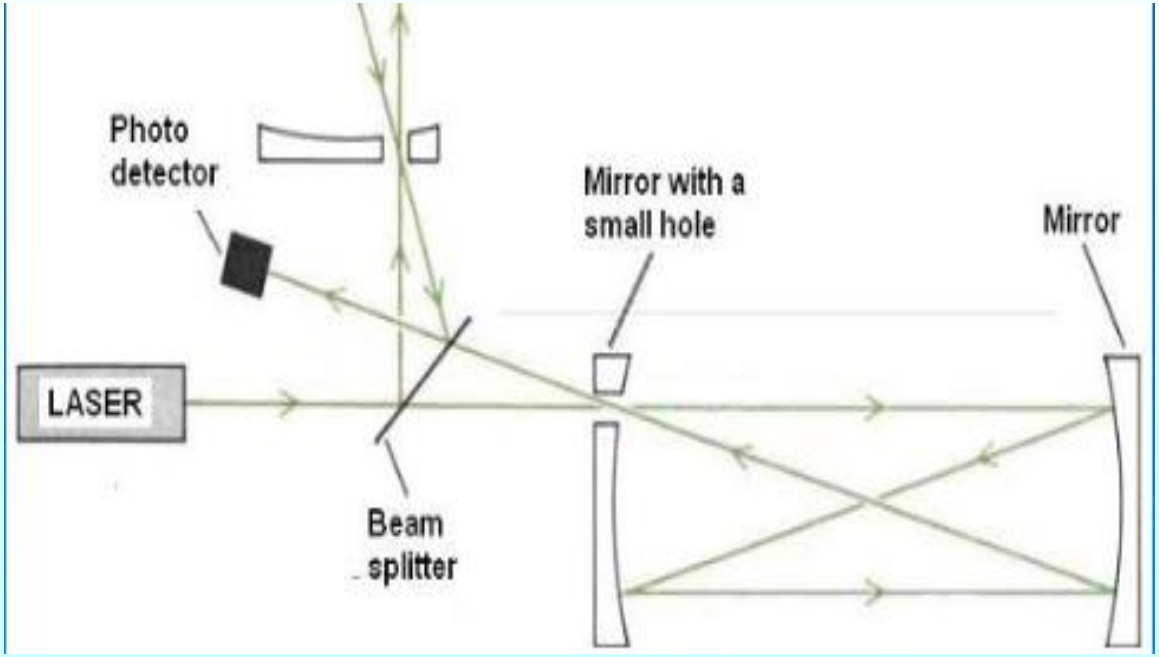
激光干涉引力波天文台（Laser Interferometer Gravitational-Wave Observatory，缩写：LIGO）是美国分别在路易斯安那州的列文斯顿和华盛顿州的汉福德建造的两个引力波探测器。



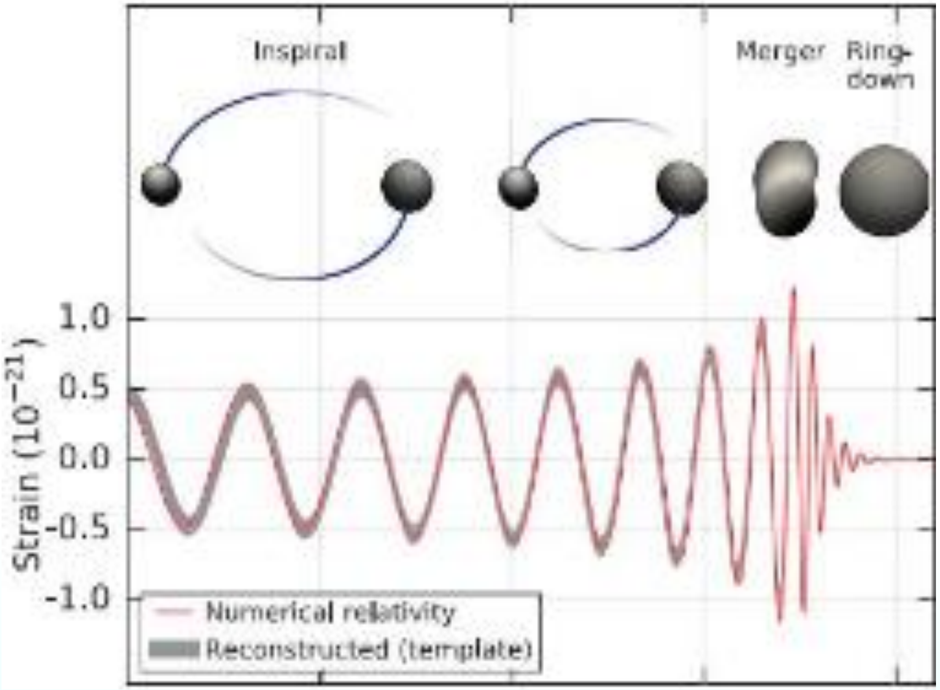
原理：垂直于干涉平面入射的引力波引起其臂长的变化，改变干涉条纹。

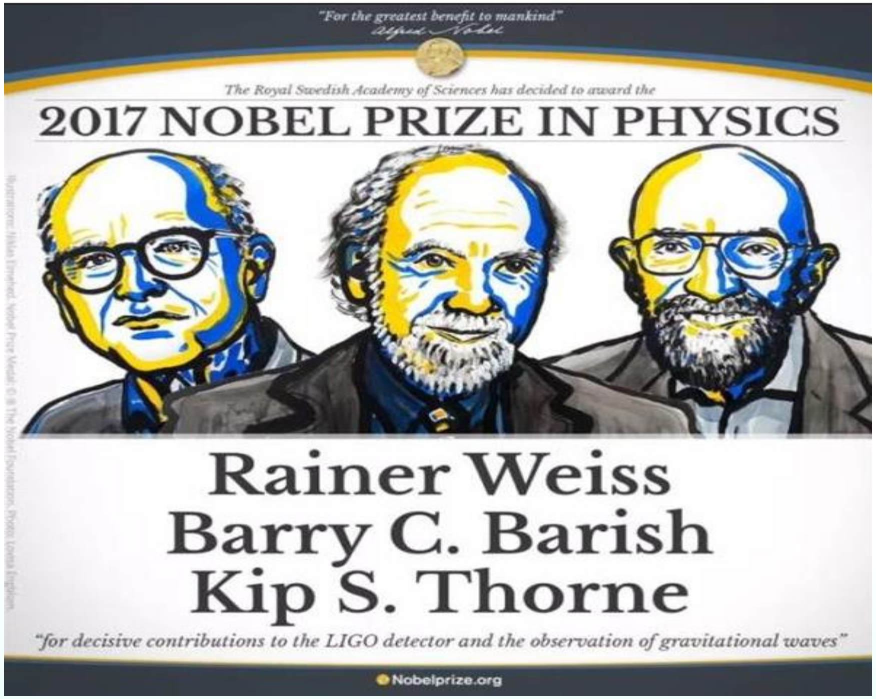
需要极高的灵敏度：
人体尺度改变了一个质子的百万分之一
日地尺度改变了一个原子
宇宙尺度改变了一个地球



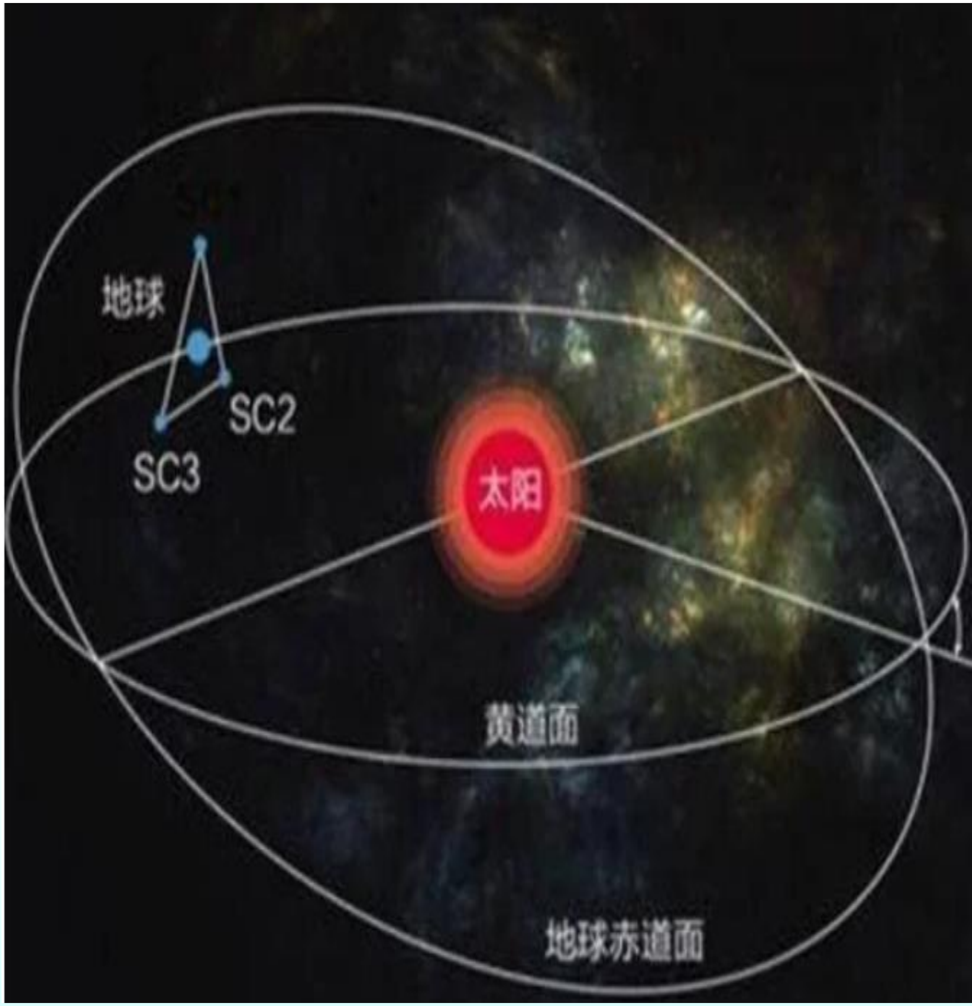


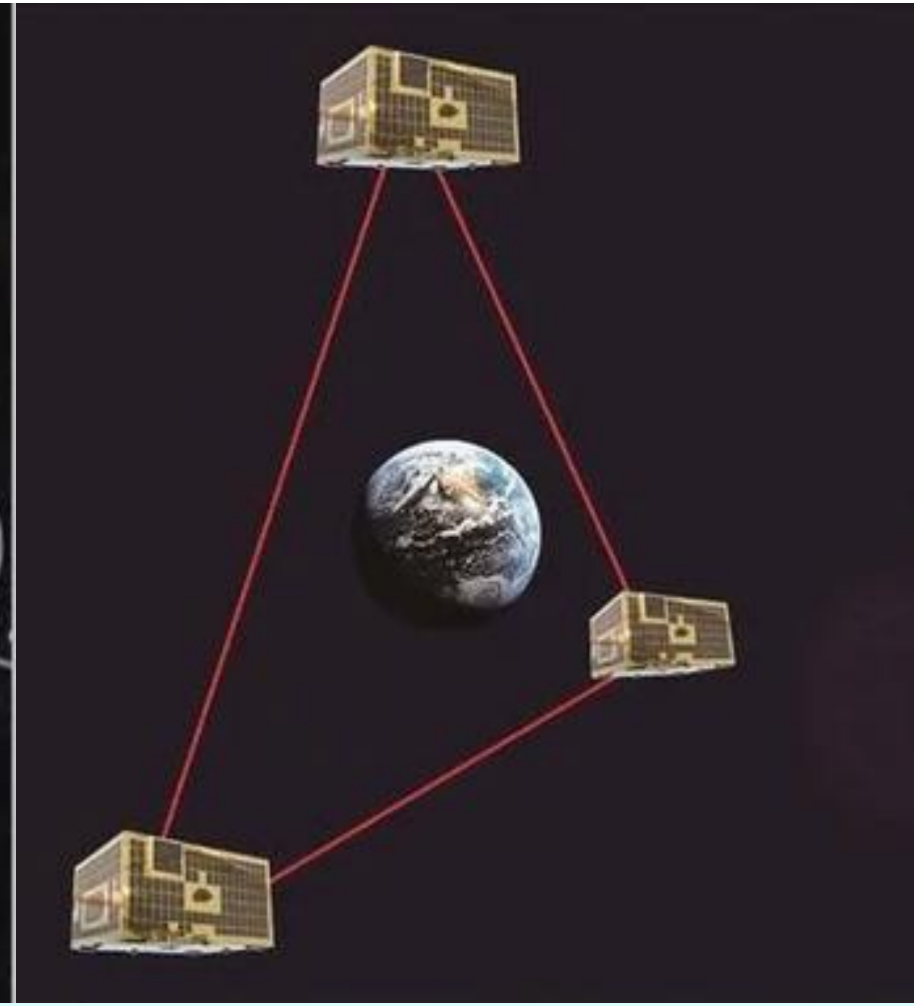
臂长：L=4000m
反射：280次
最佳频率：100-300Hz





2015年9月14日，通过LIGO计划，人类首次探测到引力波信号，由质量分别相当于29个、36个太阳的两个黑洞合并时发出。二者形成了一个62倍太阳重量的旋转黑洞，其余3个太阳质量的能量以引力波的形式释放出来。
2015年12月26日，高新激光干涉仪引力波天文台(Advanced LIGO)的两个探测器再次记录到了新的引力波信号，并合的两个黑洞质量分别相当于8个和14个太阳，形成了21倍于太阳质量的旋转黑洞。
2017年1月4日，LIGO第三次探测到引力波，来自遥远太空1.3亿年前的一次巨大撞击，长蛇座的星系NGC4993内的双中子星合并，产生引力波和电磁辐射，还接着发生了一次伽马射线暴，全球约70个地面及空间望远镜从红外、X射线、紫外和射电波段开展观测，捕捉到了来自1.3亿年前的一秒钟震动。





2019年的12月20日，长征四号火箭已经携带着“天琴一号”卫星上天了。这个“天琴一号”就是引力波探测仪之中的一个、总共需要三个类似于“天琴一号”的探测仪。在离地约十万千米高的太空轨道之中，由三个成对角的引力波探测仪来时时刻刻的观测一些激光通过时的引力波变化情况。